

A New Analysis of the Morphosyntactic–Semantic Structure of Japanese Verbal Adjectives — *KU*-adjectives¹

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Abstract:

The present paper reanalyses the morphosyntactic–semantic structure of Japanese verbal adjectives (VAs), also referred to as *i*-adjectives. Traditionally, inflected forms of this class, such as *takaku* from *takai* "to be high/tall", have been analysed as consisting of the stem *taka-* and the ending *-ku*. However, given the *multiple* morphosyntactic and semantic parallelisms with verbal stems, it is more plausible that *takaku* as a whole functions as the stem of the verbal adjective.

This paper thus analyses *taka-* as one of the two immediate constituents of the VA stem *takaku*, which encodes, as a VA radical, the lexical meaning specific to the VA, and *-ku* as the other constituent, functioning as the VA thematic — namely, stem-forming — suffix, which licences the VA radical as a syntactic constituent of the VA category at the sentential level.

The paper then examines how VAs encode the tense value [–PAST], proposing two alternative interpretations: either the phonologically null inflexional suffix *-0*, as in [[[*taka*]*i*]*0*], acquires the [–PAST] value through paradigmatic opposition to *-ta*, specified as [+PAST], as in [[[*taka*]*k+at*]*ta*]; or *-i*, as in [[[*taka*]*i*], is an amalgam of the VA thematic suffix and the non-past tense suffix. On the other hand, the atemporal form *takaku* lacks any temporal suffix, as in [[[*taka*]*ku*]]. Based on this, the paper further examines the allomorphy of the VA thematic suffix: *-i* $\approx$ *-(k)i* $\approx$ *-ku* $\approx$ *-ke*.

Finally, the paper examines the structural parallels between VAs and true verbs and the lack of surface-level parallelism between the negative forms of VAs and those of true verbs.

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1. Introduction

The inflected forms of Japanese verbal adjectives (VAs) — simply adjectives, canonical adjectives, or *i*-adjectives — are not so transparent in their morphosyntactic structure as those of verbs. In verbs, the boundary between the stem and the ending can be clearly determined. Taking the verb *taberu* "to eat" as an example, the non-past form *taberu* consists of the stem *tabe-* and the inflexional ending composed of the tense suffix *-ru*, specified as [–PAST], while the past form *tabeta* consists of the same stem *tabe-* and the inflexional ending composed of the tense suffix *-ta*, specified as [+PAST]. In other words, the opposition between the two forms is reduced to a paradigmatically transparent opposition between the suffixes *-ru* and *-ta*, both sharing the same stem *tabe-* as their inflexional base.

With VAs, on the other hand, taking *takai* "to be high/tall" as an example, the non-past form is *takai*, whereas the past form is *takakatta*. Their relationship is not as straightforward as the opposition between the non-past and past forms of verbs such as *taberu* and *tabeta*, and the boundary between the stem and the ending is, in our view, not easy to determine. Tsujimura (2007: 119) states that "adjectives are identified by a variety of conjugational endings, such as the non-past ending *-i* ... and the past tense ending *-kat-ta*." Following this, in her analysis of the VA *takai*, she treats the non-past form *takai* as consisting of the stem *taka-* + the ending *-i*, and the past form *takakatta* as the stem *taka-* + the ending *-kat-ta* (henceforth " + " with spaces on both sides indicates concatenation). This appears to represent one of the traditional and standard analyses in Japanese linguistics.

This paper, in contrast, proposes a different type of analysis, assuming that while the formation of the basic stem of verbal adjectives — the VA stem — takes place at the level of the lexicon, the combination of that basic stem and the inflexional ending occurs at the level of syntax, because such endings are assumed to constitute the phrasal head within the sentence structure. Based on this assumption, the morphosyntactic structure of verbal adjectives is analysed as in (1), where the diacritics "-" (suffixation), "()" (deletion) and "+" with no spaces on either side (compounding) — when attached immediately to morphemes — indicate the respective processes:

(1)

Stem: sentential			Ending
VA stem: lexical		Light verb stem	Inflexional suffix(es)
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-i</i>	—	<i>-0</i> [–PAST]
<i>taka-</i>	<i>-ku</i>	—	# [0 PAST]
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-ta</i> [+PAST]
<i>taka-</i>	<i>-ke</i>	—	<i>-reba</i> 'if (cond.)'

As mentioned above, the conclusive form *takai* — the so-called *syūsi-kei* in Japanese — is traditionally analysed as the stem *taka-* + the ending *-i*, but here, *takai* as a whole is treated as the lexical-level stem — that is, the VA stem. The root *taka-* is then regarded as the VA radical, which encodes the lexical meaning specific to the VA, while *-i* is analysed as the VA thematic suffix — namely, the VA stem-forming suffix. To this VA stem *takai*, the phonologically null inflexional ending *-0*, specified as [–PAST], is attached as in *takai-0*.

Similarly, the conjunctive/adverbial form *takaku* — the so-called *ren'yō-kei* — is generally analysed as the stem *taka-* + the ending *-ku*. Here, however, *takaku* as a whole is likewise treated as the VA stem, with *taka-* regarded as the VA radical and *-ku* as an allomorph of the VA thematic suffix. On the other hand, unlike the above *takai-0*, with the ending *-0* specified as [–PAST], *takaku* lacks any inflexional ending and is thus analysed as consisting solely of the stem, given its atemporality, much like the conjunctive form *tabe* from the verb *taberu*.

In the past form *takak+at-ta* — which is generally analysed as the stem *taka-* + the complex ending *-kat-ta* — the string *takak*, derived from *takaku* by final-vowel deletion, is considered the lexical-level VP stem: when the VA stem *takaku* combines with the phrasal-head tense suffix *-ta*, specified as [+PAST], at the sentential level, the stem *ar-* from the light verb *ar-u* "to be" is inserted immediately after it, as in *takaku#ar-ta*, owing to the incompatibility of the VA stem *takaku* with the verbal suffix *-ta*. This yields the final form *takak+at-ta*, with the deletion of the stem-final *-u* in *takaku* before *ar-* — which indicates the compounding of *takaku* with *ar-* — and the assimilation *ar-/ar-/* → *at-* [at-] before *-ta*.

Finally, the conditional form *takake-reba* — the so-called *katei-kei* — is generally analysed as the stem *taka-* + the ending *-kereba*. Here, however, *takake* as a whole is treated as the VA stem, with *taka-* regarded as the VA radical and *-ke* as another allomorph the VA thematic suffix, while *-reba* is exactly the same inflexional suffix found in the verbal conditional form *tabe-reba* from *taberu*. The following will examine the justification for the analysis presented in (1). The key concepts are as follows: **the multiple parallels — morphosyntactic and semantic — between the verbal stem and the conjunctive/adverbial form of VAs; the VA stem, composed of the VA radical + the VA thematic suffix $-i \approx -(k)i \approx -ku \approx -ke$; the obligatorily inserted stem $ar- \approx at- \approx ara-$ of the light verb *ar-u* "to be" in certain constructions; the phonologically zero tense suffix -0 , specified as [–PAST]; and the verbal suppletion $*ara-nai$ "not to be" → *nai*.**

2. The identification of the stem of verbal adjectives

Traditionally, as has been repeated, the conclusive form of VAs has been analysed as consisting of a stem followed by the inflexional ending *-i* — e.g. *takai* consisting of *taka-* + *-i*. On the other

hand, the conclusive form of verbs — particularly vowel-stem verbs — has been analysed as consisting of a stem followed by the inflexional ending *-ru*. In *taberu*, for instance, *tabe-* is considered the stem and *-ru* the ending. **Nevertheless, although both *taka-* and *tabe-* have been equally treated as stems, their syntactic behaviour differs fundamentally.** The verbal stem *tabe-* without overt ending can appear on its own within a sentence as in (2), whereas the so-called adjectival stem *taka-* can never occur independently in such contexts as shown in (3):

(2) *Kare =wa gohan o **tabe**, (sosite) deteitta / deteiku.* "He, having a meal, (then) left / leaves."²

(3) **Kanozyo =wa se ga **taka**, (sosite) utukusikatta / utukusii.* (ungrammatical)

"She, being tall, was / is (also) beautiful."

What can occur in the same syntactic context as the verbal stem *tabe-* is not the so-called adjectival stem *taka-* itself, but rather the conjunctive/adverbial form *takaku*, as in (4), which has traditionally been analysed — like *takai* — as consisting of the so-called stem *taka-* and the so-called inflexional ending *-ku*:

(4) *Kanozyo =wa se ga **takaku**, (sosite) utukusikatta / utukusii.* (grammatical)

"She, being tall, was / is (also) beautiful."

Accordingly, the verbal *tabe-* and the adjectival *takaku-* share the same syntactic property of being **able to occur at the end of non sentence-final coordinated clauses**. Moreover, as is clear from (2) and (4), at the stage of *Kare =wa gohan o **tabe*** and *Kanozyo =wa se ga **takaku*** in each syntactic string, **it remains indeterminate whether the utterance refers to a past or non-past situation**. Temporal specification is only achieved by the sentence-final predicate — *deteitta / utukusikatta* in the past and *deteiku / utukusii* in the non-past. Therefore, the verbal stem *tabe-* and the VA constituent *takaku-* share the same semantic property of being **atemporal**: they do not themselves make any temporal reference, indicating that they both **lack inflexional tense endings** — a **defining feature of the predicative stem**. Furthermore, the similarity between the verbal stem *tabe-* and the VA constituent *takaku-* is not limited to these points; as shown in (5) – (10), both **can appear in the sentence, followed by the inflexional suffixes *-te*, *-temo* and *-tatte***.

(5) *Kare =wa gohan o **tabe-te** deteitta / deteiku.* "He, having a meal, then left / leaves."

(6) *Kanozyo =wa se ga **takaku-te** utukusikatta / utukusii.* "She, being tall, was / is also beautiful."

² The translations of the examples in this paper are given in a *literal* manner and may therefore sound — sometimes very — unnatural in English.

- (7) *Kare* =wa gohan o *tabe-temo* hutoranai. "He, even having a meal, does not gain weight."
 (8) *Kanozyo* =wa se ga *takaku-temo* medatanai. "She, even being tall, does not stand out."
 (9) *Kare* =wa gohan o *tabe-tatte* hutoranai. "He, even if he eats, does not gain weight."
 (10) *Kanozyo* =wa se ga *takaku-tatte* medatanai. "She, even if tall, does not stand out."

In contrast, the forms **taka-te*, **taka-temo* and **taka-tatte* — each composed of the so-called stem *taka-* followed by an inflexional suffix *-te*, *-temo* and *-tatte* — are all ungrammatical. In this way, there is clear syntactic, semantic and morphosyntactic parallelism between *tabe-* and *takaku-*. Given that *tabe-* is analysed as the stem of the verb, it seems more appropriate to treat *takaku-* — not as the concatenation of the so-called stem *taka-* and the so-called inflexional ending *-ku*, as traditionally assumed — but rather as the stem of the verbal adjective in its entirety, which we shall henceforth refer to as the VA stem.

Historically, there was an etymological form in Old Japanese from which the modern verbal form *tabe-tatte* originated — namely, *tabe-tari-tote*, which later changed to *tabe-ta-tote*. In contrast, the modern VA form *takaku-tatte* does not have a comparable direct etymon with the same structure in Old Japanese, such as **takaku-tari-tote*, nor in later Japanese, such as **takaku-ta-tote* — both of which would have been agrammatical. What this suggests is that the modern grammatical form *takaku-tatte* seems to have been formed by inserting the VA stem *takaku-* into the verbal stem position within the already existing word structure composed of "the verbal stem + the inflexional suffix *-tatte*" — as in *tabe-tatte* — on the basis of their morphosyntactic commonality as predicative stems. If *taka-* were, like *tabe-*, a predicative stem, it would be impossible to explain why **taka-tatte* — obtained by replacing the genuine stem *tabe-* with the so-called stem *taka-* in *tabe-tatte* — results in ungrammaticality. In other words, **the very existence of such an inflected form as *takaku-tatte* can be taken as evidence that the VA stem, which has the same status as verbal stems such as *tabe-* as predicative stems, is not *taka-* but *takaku-***. All of the above discussion leads to the following question: how, then, should *taka-* — traditionally regarded as the stem — and *-ku* — traditionally regarded as the inflexional ending — be reanalysed as morphological constituents? This issue will be discussed in the following section.

3. The radical and the thematic suffix of verbal adjectives

When we observe all the inflected forms of the VA *takai* — namely, *takai* $\approx$ *taka(k)i* $\approx$ *takaku* $\approx$ *takakute* $\approx$ *takakutemo* $\approx$ *takakutatte* $\approx$ *takakatta* $\approx$ *takakattarō* $\approx$ *takakattara(ba)* $\approx$ *takakattari* $\approx$ *takakarō* $\approx$ *takakare* $\approx$ *takakereba* — we find that they all share the element *taka-*. This *taka-* is

also attested in derived nouns such as *takasa* "height" and *takami* "loftiness" as well as in compound nouns such as *takanozomi* "high hope" and *takadai* "high platform", where it refers to the very property or state of "being high/tall". Since it cannot be further decomposed into smaller morphemes, it is to be regarded as a root. However, as the ungrammaticality of (11) clearly shows, the root *taka-* alone cannot function as a verbal adjective — that is, it cannot be inserted into a terminal node dominated by the VA node — and thus cannot establish structural relations with other constituents within the sentence:

(11) **Koko kara, taka sobieru tō ga mieru.* (ungrammatical)

"From here, a tower rising high can be seen."

By contrast, example (12), in which *taka-* is accompanied by *-ku*, is grammatical:

(12) *Koko kara, takaku sobieru tō ga mieru.* (grammatical)³

"From here, a tower rising high can be seen."

Now, as the derived words and compound words cited above show, **the root *taka-* functions only as a mere constituent of word formation**: it cannot itself serve as a syntactic unit at the sentential level. We are thus led to posit that *taka-* requires an additional element in order to function as a syntactic category VA at the sentential level. This element is precisely *-ku*, which is present in the sentential form *takaku*, but absent from word-level forms such as *takasa*, *takami*, *takanozomi* and *takadai*. As observed in Section 2, the addition of *-ku* to the root *taka-* yields a verbal adjective stem that is functionally — more precisely, morphosyntactically and semantically — equivalent to the verbal stem *tabe-*. We shall therefore refer to *-ku* as the **VA thematic suffix** — namely, the **VA stem-forming suffix**. (From the perspective of Distributed Morphology, this constituent may be regarded as an *overt* instantiation of the categorial head *a*, a category-defining element that is otherwise often assumed to be *covert*.)

The historical development of verbal adjectives also supports this interpretation: "Originally, adjectives, remaining as stems, were positioned before nouns and verbs, modifying them, as in *nagauta*, *takayuku*. However, to explicitly express the functions of the conjunctive, attributive and conclusive forms, inflected forms such as *takaku*, *takaki*, *takasi* were established. Moreover, after the Nara period, forms such as *takakere* developed as realis forms *izenkei*. In this way, adjectives performing the role of predicates developed more slowly" (Ōno, S., et al. (Eds.). 1990:

³ Note with respect to this adverbial use of VA phrases (VAPs) that, in this study, it is assumed that **a phonologically zero adverbial phrase head (AdvP head) *-0* merges with VAPs** — e.g. *takaku* — **to form AdvPs** — e.g. *takaku-0* — **which function as adjuncts of predicates**.

1467, translated from Japanese). From the perspective of this paper, **the root of VAs** — e.g. *naga-* "long" and *taka-* "high/tall" — **originally functioned only as a formative element at the lexicon level, forming compounds** such as *naga+uta* "long poem" and *taka+yuku* "to fly high into the sky", **placed before base nouns or base verbs. However, in order to function as the head of a VA phrase at the sentential level, that is, to function as a category VA, the thematic suffix** *-ku* $\approx$ *-ki* $\approx$ *-si* $\approx$ *-ke*, as in *taka-ku* $\approx$ *taka-ki* $\approx$ *taka-si* $\approx$ *taka-ke* — **corresponding to** *-ku* $\approx$ *-(k)i* $\approx$ *-i* $\approx$ *-ke* **in Contemporary Japanese**, as in *taka-ku* $\approx$ *taka-(k)i* $\approx$ *taka-i* $\approx$ *taka-ke* — **was added**. The VA thematic suffix proposed in this paper corresponds to what Frellesvig (2010: 80–90, *inter alia*) refers to as the "adjectival copula", although its allomorphs *-ku* $\approx$ *-ki* $\approx$ *-si* $\approx$ *-ke* constitute part of his "Old Japanese adjectival copula forms" (listed in Table 3.8, *ibid.*: 81). The origin of these conjunctive/adverbial form, attributive form, conclusive form and realis form ending in the VA thematic suffix *-ku* $\approx$ *-ki* $\approx$ *-si* $\approx$ *-ke* — namely, *ren'yōkei*, *rentaikei*, *shūshikei* and *izenkei* — is explained in Okimori (2017: 85–88).

Let us now turn to the morphological status of the constituent *taka-* within the VA stem. In the preceding discussion, *taka-* was referred to as a *root* in relation to *-ku*, but this designation is not strictly accurate from the viewpoint of morphological structure. As the constituent *na+daka-* — derived from *na* "name" + *taka-* "high", with *rendaku* in the second element — in the compound verbal adjective *na+daka-i* "to be famous" in (13) shows, what precedes the VA thematic suffix need not be a single root, but may also involve root-root compounding:

(13) *Yamazaki =wa na+daka-i gengogakusya da.* "Yamazaki is a famous linguist."

It is therefore morphologically inappropriate to refer to the constituent preceding the VA thematic suffix *-ku* $\approx$ *-i* as a root. In what follows, we shall refer to it as a **VA radical**, in the sense of **one of the two immediate constituents of a VA stem that constitutes its semantic core**.

As the inflected forms *takaku*, *takaku-te*, *takaku-temo* and *takaku-tatte* in (4), (6), (8) and (10) show, the VA stem formed with the suffix *-ku* — namely, *takaku-* — can appear on its own, or followed by the suffixes *-te*, *-temo* and *-tatte*. From this, the structure of verbal adjectives may be analysed as in (14):

(14)

VA stem		Ending
VA radical	VA thematic suffix	Inflexional suffix
<i>taka-</i>	<i>-ku</i>	#
<i>taka-</i>	<i>-ku</i>	<i>-te</i>
<i>taka-</i>	<i>-ku</i>	<i>-temo</i>
<i>taka-</i>	<i>-ku</i>	<i>-tatte</i>

It can be confirmed that the morphemes *-te*, *-temo* and *-tatte* are inflexional suffixes as follows. In *takaku-te*, *takaku-temo* and *takaku-tatte*, it is possible to insert syntactically the focus particle *=mo* "too, also" or the topicalisation/contrast particle *=wa* "as for, etc." — both of which are enclitics — immediately after the verbal adjective stem *takaku-*. In these cases, the stem allomorph *at-* from the light verb *ar-u* "to be" is obligatorily inserted before *-te*, *-temo* and *-tatte*, as shown in (15), where "#" represents a morphological word boundary that coincides with a prosodic word boundary and "=" indicates enclisis:

- (15) *takaku-te* → *takaku=mo#at-te* vs. **takaku=mo#te* "being also high/tall"
takaku-temo → *takaku=wa#at-temo* vs. **takaku=wa#temo* "high/tall, though it is so"
takaku-tatte → *takaku=wa#at-tatte* vs. **takaku=wa#tatte* "high/tall, even if it is so"

In the case of the verbal inflected forms *tabe-te*, *tabe-temo* and *tabe-tatte* in (5), (7), (9), the situation is exactly the same, as indicated in (16):

- (16) *tabe-te* → *tabe=mo#si-te* vs. **tabe=mo#te* "doing also eat"
tabe-temo → *tabe=wa#si-temo* vs. **tabe=wa#temo* "though it does eat"
tabe-tatte → *tabe=wa#si-tatte* vs. **tabe=wa#tatte* "even if it does eat"

In this case, a stem allomorph *si-* from the light verb *su-ru* "to do" is inserted before *-te*, *-temo* and *-tatte*. From (15) and (16), the following observations (17) – (20) may be drawn:

(17) In *takaku-te*, *takaku-temo* and *takaku-tatte* in (15) and *tabe-te*, *tabe-temo* and *tabe-tatte* in (16), respectively, it is possible to insert the enclitic particles *=mo* or *=wa* immediately after the VA stem *takaku-* and the verbal stem *tabe-* — henceforth, V stem — at the syntactic level. This indicates that *takaku-* and *tabe-* can occupy the same structural position in syntax and, here too, **syntactic parallelism is observed between *takaku-* and *tabe-***.

(18) In both (15) and (16), if *-te*, *-temo* and *-tatte* were free morphemes and could constitute autonomous words, then forms such as **takaku=mo#te*, etc. and **tabe=mo#te*, etc. should be grammatical as they are. This is not the case, however: only when the stems *at-* and *si-* of the light verbs *ar-u* and *su-ru* are inserted, respectively, do the forms become grammatically adequate. It follows, then, that *-te*, *-temo* and *-tatte* are inflexional suffixes which cannot constitute a word independently and, formally, always require a stem.

(19) Based on (18), the insertion of the light verb stems *at-* and *si-* in *takaku-te* → *takaku=mo#at-te*, etc. in (15) and *tabe-te* → *tabe=mo#si-te*, etc. in (16) indicates that, since both *takaku-* and *tabe-* function as stems for the inflexional suffixes *-te*, *-temo* and *-tatte* in *takaku-te*, etc. and *tabe-te*, etc., the insertion of the enclitic particles *=mo* or *=wa* between the stems and the suffixes renders the latter stemless, and therefore requires the subsequent insertion of alternative formal stems — namely, *at-* and *si-* — before them, because *=mo* and *=wa* cannot function as stems for them. This again supports the claim that **it is not *taka-* but *takaku-* in its entirety that functions as the stem in VAs** and, here too, **morphosyntactic parallelism is observed between *takaku-* and *tabe-***.

(20) In (15), the reason why it is not some VA stem but the V stem *at-* that is inserted in *takaku=mo#at-te*, etc. is simply that Japanese lacks a class of light verbal adjectives and their corresponding stems, and hence some V stem must be instead employed to support formally the suffixes *-te*, *-temo* and *-tatte* even in *takaku=mo#at-te*, *takaku=wa#at-temo* and *takaku=wa#at-tatte*. But why is it not the stem *si-* from *su-ru* but the stem *at-* from *ar-u* that is selected? The motivation seems to be that the prototypical categorical distinction — *stativity* vs. *dynamicity* — between the VA stem *takaku-* and the V stem *tabe-* conditions the selection of stems between the *stative* light verb *ar-u* "to be" and the *dynamic* light verb *su-ru* "to do". The syntactic process that determines this selection based on semantics remains a topic for future investigation.

Together with the description at the end of Section 2, it can be observed that **between the VA stem *takaku-* and the V stem *tabe-*, clear and multiple parallelism — morphosyntactic and semantic — is evident**.

Next, let us turn to more complex structures: in the inflected forms *taka-k+at-ta*, *taka-k+at-tar-ō*, *taka-k+at-tar-a(ba)*, *taka-k+at-tar-i*, *taka-k+ar-ō*, *taka-k+ar-e*, if the focus particle *=mo* "too, also" is inserted before the conjugated forms of the light verb *ar-u* "to be" — namely, *at-ta*, *at-tar-ō*, *at-tar-a(ba)*, *at-tar-i*, *ar-ō* and *ar-e* — and the string is thereby divided into two words, the following alternations in (21) can be observed:

(21) <i>taka-k+at-ta</i> ≈ <i>taka-<u>ku</u>=mo#<u>at-ta</u></i>	"It was high/tall" ≈ "... also high/tall"
<i>taka-k+at-tar-ō</i> ≈ <i>taka-<u>ku</u>=mo#<u>at-tar-ō</u></i>	"It would be high/tall" ≈ "... also high/tall"
<i>taka-k+at-tar-a(ba)</i> ≈ <i>taka-<u>ku</u>=mo#<u>at-tar-a(ba)</u></i>	"If it is/were high/tall" ≈ "... also high/tall"
<i>taka-k+at-tar-i</i> ≈ <i>taka-<u>ku</u>=mo#<u>at-tar-i</u></i>	"Sometimes it is high/tall" ≈ "... also high/tall"
<i>taka-k+ar-ō</i> ≈ <i>taka-<u>ku</u>=mo#<u>ar-ō</u></i>	"It will be high/tall" ≈ "... also high/tall"

taka-k+ar-e $\approx$ *taka-ku=mo#ar-e*

"Be high/tall" $\approx$ "... also high/tall"

From this, it can be seen that the VA thematic suffix *-ku* undergoes vowel deletion and surfaces as *-k* when it precedes the vowel-initial stem of the light verb *ar-u*, namely *ar-* $\approx$ *at-*. This vowel deletion may be taken to indicate that the VA stem and the light verb stem — each of which can function as an autonomous word — have come to form a single compound stem.

The reason why the inflected forms in (21) contain *ar-* or *at-* — stem allomorphs of the light verb *ar-u* — is that the endings *-ta*, *-tar-ō*, *-tar-a(ba)*, *-tar-i*, *-ō*, *-e*, which encode tense or/and modality, combine only with purely verbal stems. In other words, *takaku* is a non-verbal stem and thus cannot combine directly with these endings. Consequently, the VA stem *taka-ku* combines with the stem *ar-* of the light verb *ar-u*, forming a **compound V stem *taka-k+ar-*** ($\leftarrow$ *taka-ku* + *ar-*) at the sentential level, which as a whole comes to **possess verbal status and be able to combine with the verbal endings *-ta*, *-tar-ō*, *-tar-a(ba)*, *-tar-i*, *-ō*, *-e***. This is made possible by the fact that Japanese is a head-final language, whereby the light verb stem *ar-* determines the syntactic category of the entire compound *taka-k+ar-*. Under this interpretation, the structure of (14) may be more accurately represented as in (22), where "-", "()" and "+" indicate suffixation, deletion and compounding, respectively, as in (1):

(22)

Stem: sentential			Ending
VA stem: lexical		Light verb stem	Inflexional suffix(es)
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-ku</i>	—	#
<i>taka-</i>	<i>-ku</i>	—	<i>-te</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-temo</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-tatte</i>
<i>taka-</i>	<i>-k(u)</i>	+ <i>at-</i>	<i>-ta</i>
<i>taka-</i>	<i>-k(u)</i>	+ <i>at-</i>	<i>-tar-ō</i>
<i>taka-</i>	<i>-k(u)</i>	+ <i>at-</i>	<i>-tar-a(ba)</i>
<i>taka-</i>	<i>-k(u)</i>	+ <i>at-</i>	<i>-tar-i</i>
<i>taka-</i>	<i>-k(u)</i>	+ <i>ar-</i>	<i>-ō</i>
<i>taka-</i>	<i>-k(u)</i>	+ <i>ar-</i>	<i>-e</i>

Besides the both morphosyntactically and phonologically conditioned allomorphy *-ku* $\approx$ *-k*, the VA thematic suffix exhibits another type of allomorphy *-ku* $\approx$ *-i* $\approx$ *-(k)i* $\approx$ *-ke* depending on the syntactic environment in which it occurs, as shown in (23) – (28):

(23) **Kanozyo =wa se ga taka.*

(24) *Kanozyo =wa se ga takai.* "She is tall."

(25) **Asoko ni se no taka otoko ga iru.*

(26) *Asoko ni se no takai otoko ga iru.* (Classical: *takaki*) "There is a tall man over there."

(27) **Se ga taka-reba ii to yū wake de =wa nai.*

(28) *Se ga takake-reba ii to yū wake de =wa nai.* "It is not always the case that it is good if the height is tall."

Among these, *taka-ke* in (28), which bears the allomorph *-ke*, occurs in modern Japanese exclusively before the suffix *-reba*, which signifies a sequential conditional: hence, *taka-ke-reba* is referred to as *kateikei* — the conditional form. By contrast, the syntactic conditions that determine the occurrence of *taka-i* in (24) and *taka-(k)i* in (26), bearing the allomorphs *-i* and *-(k)i* respectively, will be examined in the following section.

4. Syntactic conditions governing the allomorphy: $-i \approx -(k)i \approx -ku \approx -ke$

As observed in (4), the conjunctive/adverbial form *taka-ku*, in which the VA thematic suffix appears in the form *-ku* and no inflexional ending follows, is **atemporal**. The fact that the form *taka-ku* lacks temporal reference is consistent with the interpretation that it does not bear any inflexional suffix encoding tense or/and modality. By contrast, the conclusive form *taka-i* and the attributive form *taka-(k)i* — the so-called *rentaikei* — appear to lack any inflexional ending just as the conjunctive/adverbial form *taka-ku* does, yet unlike *taka-ku*, they bear **non-past value**. The question, then, is how this non-past value is formally encoded.

Traditionally, the final *-i* in *taka-i* and *taka-(k)i* has often been interpreted as the word constituent that bears the temporal value of non-past. This interpretation is difficult to justify, however, both historically and in light of the classical yet fundamental thesis that **linguistic value arises from opposition**, as Saussure (1916: 187; 196) puts it: "Dans l'intérieur d'une même langue, tous les mots qui expriment des idées voisines se limitent réciproquement ... n'ont de valeur propre que par leur opposition"; "deux signes comportant chacun un signifié et un signifiant ... sont seulement distincts. Entre eux il n'y a qu'opposition. Tout le mécanisme du langage ... repose sur des oppositions de ce genre et sur les différences phoniques et conceptuelles qu'elles impliquent".

For the *-i* in *taka-i* to acquire the [–past] value, it must be paradigmatically opposed to the past suffix *-ta*. This would require the existence of a form **taka-ta*, in the same way that the suffix *-ru* in *tabe-ru* acquires the [–past] value only by being opposed to the suffix *-ta* in *tabe-ta*, but in fact no such form as **taka-ta* exists. Conversely, for the *-ta* in the attested form *takakat-ta* to acquire the [+past] value in opposition to *-i*, the hypothetical form **takakat-i* would be required. To what, then, is the *-i* in *taka-i* actually opposed in order to acquire the [–past] value?

Historically, verbal adjectives originally lacked a non-past vs. past tense opposition and, accordingly, they contained no formal element that encoded tense. For example, in Old Japanese of the 8th century, forms such as *taka-si*, *taka-ki* and *taka-ku* consisted, from the standpoint of the present analysis, solely of the stem, lacking any inflexional ending. However, probably due to the increasing need to encode the *past* value more explicitly, the past tense suffix *-ki* — which was solely compatible with verbal stems — and the stem allomorph *ari-* with epenthetic *-i* of the light verb *ari* "to be" were employed to constitute the past tense form *taka-k+ari-ki*. As Okimori (2017: 84) states, "The secondary conjugation known as *kari*-conjugation was formed by the conjunctive/ adverbial form being followed by the *r*-irregular verb *ari*, through the development *ku-ari* > *kari*. This made it possible for the form to attach to a wide range of particles and auxiliary verbs" (translated from Japanese). Frellesvig (2010: 90) likewise notes: "In addition to its basic inflected forms, the adjectival copula formed extended analytic forms by the infinitive *-ku* followed by *ar*...: *-ku ar*-. ... This allowed the formation of ... combination with auxiliaries which did not combine directly with the adjectival copula.... The analytic formations provided full morphophonological versatility for the adjectival copula. The analytic forms sometimes fused phonologically, *-ku ar*- => *-kar*-, which in EMJ gave rise to a secondary conjugation of the adjectival copula".

This suggests that the conclusive form *taka-si* and the attributive form *taka-ki*, which originally lacked any overt tense marker, came to bear *non-past* value through their formal opposition to the newly constructed past form *taka-k+ari-ki* — that is, by lacking the overt tense suffix *-ki*. In other words, from the perspective of formal opposition, those two forms came to be analysed as bearing a **phonologically null inflexional ending -0 specified as [-PAST]** as in [[[*taka*]*si*]*0*] and [[[*taka*]*ki*]*0*], which came to stand in opposition to the past-tense inflexional suffix *-ki* specified as [+PAST] as in [[[*taka*]*k+ari*]*ki*], which is consistent with Saussurean view that "La langue est un système de pures valeurs que rien ne détermine en dehors de l'état momentané de ses termes" (Saussure 1916: 132). This is parallel to the case of English *cat-0*, where the lack of an overt plural marker in opposition to *cat-s* functions to mark the singular. Based on the above, the morphosyntactic structure of the classical forms *taka-si*, *taka-ki*, *taka-ku* and *taka-k+ari-ki* may be analysed as in (29):

(29)

Stem: sentential			Ending
VA stem: lexical		Light verb stem	Inflexional suffix
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-si</i>	—	-0 [-PAST]
<i>taka-</i>	<i>-ki</i>	—	-0 [-PAST]
<i>taka-</i>	<i>-ku</i>	—	# [0PAST]
<i>taka-</i>	<i>-k(u)</i>	+ari-	-ki [+PAST]

In the case of modern Japanese as well, it may be assumed that the partial system in (29) remains unchanged, with the past tense suffix *-ki* simply having been replaced by *-ta*, as shown in (30):

(30)

Stem: sentential		Ending	
VA stem: lexical		Light verb stem	Inflexional suffix
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-i</i>	—	<i>-0</i> [−PAST]
<i>taka-</i>	<i>-(k)i</i>	—	<i>-0</i> [−PAST]
<i>taka-</i>	<i>-ku</i>	—	# [0PAST]
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-ta</i> [+PAST]

What remains to be discussed are the relationship between the conclusive form *taka-i* and the attributive form *taka-(k)i*, and the conditions that determine their distribution in relation to the conditional form *taka-ke* and the conjunctive/adverbial form *taka-ku*. The first two forms, *taka-i* and *taka-(k)i*, share the VA radical *taka-* and a zero tense suffix *-0* specified as [−PAST], but differ in the shape of the VA thematic suffix: *-i* in the conclusive form and *-(k)i* in the attributive form. Correspondingly, they occur in distinct syntactic environments:

(31) *Utukusi-(k)i hito* "A beautiful person":

[_{NP} [_{CP} [_{TP} [_{VAP} op_i *utukusi-(k)i*] -0] -0] *hito*_i] → [_{NP} [_{CP} [_{TP} [_{VAP} op_i *utukusi-(k)i*] -0] -0] *hito*_i]⁴

(32) *Kanozyo=wa utukusi-i* "She is beautiful":

[_{TopP} [_{TP} [_{VAP} *kanozyo utukusi-i*] -0] -0] → [_{TopP} *kanozyo*_i=*wa* [_{TP} [_{VAP} *t_i utukusi-i*] -0] -0]

As shown in (31), when an VAP functions as the complement of TP headed by T -0 [−PAST] and this TP is embedded within an NP, the VA thematic suffix appears in the form *-(k)i*. By contrast, when an VAP functions as the complement of TP headed likewise by T -0 [−PAST] but the TP is not embedded within an NP, as in (32), the VA thematic suffix appears in the form *-i*. As for the remaining forms *-ke* and *-ku*, *-ke* in modern Japanese is exclusively distributed in the environment preceding the TP-head suffix *-reba* as in (33):

(33) *Kanozyo ga utukusi-ke-reba* "If she is beautiful":

[_{TP} [_{AP} *kanozyo utukusi-ke*] -reba] → [_{TP} *kanozyo*_i=*ga* [_{AP} *t_i utukusi-ke*] -reba]

⁴ There is also the possibility of [_{NP} [_{CP} [_{TP} [_{VAP} op_i *utukusi-(k)i*] -0] -0] *hito*_i] → [_{NP} [_{CP} op_i [_{TP} [_{VAP} *t_i utukusi-(k)i*] -0] -0] *hito*_i]. However, this paper employs a simplified model focusing on the allomorphy of the VA thematic suffix *-i* ≈ *-(k)i* ≈ *-ku* ≈ *-ke* and on its syntactic conditions, leaving full syntactic verification to future work.

while *-ku* is the form taken by the VA thematic suffix in all other syntactic environments, namely **the elsewhere form**, when the conditions for the more specific forms *-i*, *-(k)i*, *-ke* are not met. Thus, it seems **more appropriate to refer to VAs as *ku*-adjectives** rather than *i*-adjectives, after their **default form in *-ku*** — diachronically the earliest form in which VAs were used as syntactic constituents at the sentential level, as Okimori (2017: 85–86, translated from Japanese) states: "Adjectives originally had as their basic form only the conjunctive form used adverbially. The conjunctive inflexional ending *-ku* is ... an adverbial ending and the conjunctive form was originally an adverb. ... That adverb came to function like a predicate and came to have the use of the suspensive form." Finally, on this paper's view that the formation of VA stems is effected in the lexicon, the syntactic derivations in (31) – (33) and the allomorphy of the VA thematic suffix *-ku* can, more precisely, be represented as in (34):

$$\begin{aligned}
 (34) \quad & [\text{NP} [\text{CP} [\text{TP} [\text{VAP op}_i \text{ utukusi-ku}] -0] -0] \text{ hito}_i] \rightarrow [\text{NP} [\text{CP} [\text{TP} [\text{VAP op}_i \text{ utukusi-(k)i}] -0] -0] \text{ hito}_i] \\
 & [\text{TopP} [\text{TP} [\text{VAP kanozyo utukusi-ku}] -0] -0] \rightarrow [\text{TopP} \text{ kanozyo}_i=\text{wa} [\text{TP} [\text{VAP } t_i \text{ utukusi-i}] -0] -0] \\
 & [\text{TP} [\text{AP kanozyo utukusi-ku}] -reba] \rightarrow [\text{TP} \text{ kanozyo}_i=\text{ga} [\text{AP } t_i \text{ utukusi-ke}] -reba]
 \end{aligned}$$

5. An alternative interpretation of the allomorphy $-i \approx -(k)i \approx -ku \approx -ke$

Building on the structuralist–functionalist notion of *amalgam*, another type of analysis for the allomorphy $-i \approx -(k)i \approx -ku \approx -ke$ becomes possible. Regarding the notion of amalgam — *amalgame* in French — Martinet (1991: 102) writes: "deux signifiés qui coexistent dans un énoncé enchevêtrent leurs signifiants de telle façon qu'on ne saurait analyser le résultat en segments successifs".

As is clear from the description in the final paragraph of Section 4, the allomorph *-ku* of the VA thematic suffix, unlike *-i*, and *-(k)i*, which occur under specific morphosyntactic conditions as in (31) – (33), can be regarded as the default form — namely, the form that surfaces when no specific morphosyntactic condition is met, or the *elsewhere* form. Furthermore, whereas *-ku* can occur on its own without any following morpheme, *-i* and *-(k)i* always appear with a specific morpheme following them — namely, *-0* [–PAST]. From this, the amalgams in (35) become conceivable:

$$(35) \quad -ku + [-\text{PAST}] \rightarrow -i \text{ or } -(k)i, \text{ depending on morphosyntactic conditions.}$$

Under this analysis, ***-i* and *-(k)i* are, synchronically, not allomorphs exclusively of the VA thematic suffixes** but **simultaneously encode the value [–PAST]**, thereby *partially* overlapping with the traditional analysis. It should nevertheless be noted that, as indicated in Section 4, ***-i* and**

-(k)i cannot be interpreted as allomorphs exclusively of the tense suffix [-PAST] either, in light of the Saussurean thesis that linguistic value arises from opposition.

6. Structural similarities with verbal inflected forms

The following examines the inflected forms of the vowel-stem verb *nobiru* "to stretch/extend" (intransitive) and its classical form *nobu*, and demonstrates the significant structural parallels between these forms and the inflected forms of the verbal adjective *takai* "to be high/tall" and its classical form *takasi*.

The one-argument verb *nobiru* is a non-accusative one, with the stem *nobi-*. On the other hand, the morphosyntactically and semantically corresponding two-argument and accusative verb *noberu* "to stretch/extend" (transitive) has the stem *nobe-*. The shared element *nob-* in both the non-accusative verb stem *nobi-* and the accusative verb stem *nobe-* is present in all the inflected forms of these two verbs. Moreover, since the meaning of "stretching/extension" as a state change is common to all their inflected forms, it can be concluded that *nob-* represents the state change of "stretching/extension". In modern Japanese, although the accusative verb corresponding to *nobiru* is more commonly *nobasu* — which can be analyzed as underlyingly *nob-sas-ru* with the causative morpheme *-sas-*, surfacing as *nob-as-u* — the stem *nobas-* itself also contains *nob-*, representing the same state change.

Thus, the stem *nobi-* contains a morpheme boundary within it, as in *nob-i-*. Since neither *nob-* nor *-i* can be further segmented into smaller sequential units, *nob-* is the immediate constituent of the verbal stem *nob-i-* and, encoding the core lexical meaning of the verb, can be considered, alongside the VA radical discussed in Section 3, as the radical of the verb — the **V radical**. On the other hand, the *-i* in *nob-i-* combines with the V radical *nob-* to form the stem of the verb — the **V stem** — and it is assumed to function to licence the V radical as a category V at the sentential level. Thus, in parallel with the VA thematic suffix discussed in Section 3, *-i* can be considered the thematic suffix of the verb — the **V thematic suffix**. This suffix *-i* is also assumed to confer non-accusativity and a one-argument structure as its valency to the V radical *nob-* through its opposition to *-e* in the accusative verb stem *nob-e-*, which, like *-i*, is a V thematic suffix.

Based on the above analysis, the morphosyntactic structure of the following inflected forms of VA *takai* — *taka-i-0*, *taka-(k)i-0*, *taka-ku*, *taka-ke-reba*, *taka-k+at-ta* — and their corresponding forms of V *nobiru* — *nob-i-ru*, *nob-i-ru*, *nob-i*, *nob-i-reba*, *nob-i-ta* — is as shown in (36):

(36)

Stem: sentential			Ending
Stem: lexical		Light verb stem	Inflexional suffix
Radical	Thematic suffix		
<i>taka-</i>	<i>-i</i>	—	<i>-0</i> [–PAST]
<i>nob-</i>	<i>-i</i>	—	<i>-ru</i> [–PAST]
<i>taka-</i>	<i>-(k)i</i>	—	<i>-0</i> [–PAST]
<i>nob-</i>	<i>-i</i>	—	<i>-ru</i> [–PAST]
<i>taka-</i>	<i>-ku</i>	—	# [0 PAST]
<i>nob-</i>	<i>-i</i>	—	# [0 PAST]
<i>taka-</i>	<i>-ke</i>	—	<i>-reba</i>
<i>nob-</i>	<i>-i</i>	—	<i>-reba</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-ta</i> [+PAST]
<i>nob-</i>	<i>-i</i>	—	<i>-ta</i> [+PAST]

Except that the tense [–PAST] is encoded by *-0* in VA *taka-i-0* and by *-ru* in V *nob-i-ru*, and that, in VA, the stem *at-* of the light verb *ar-u* may be inserted at the sentential level, the basic structure is the same. The inflected forms of the corresponding classical forms *takasi* and *nobu* exhibit even greater structural similarity, as indicated in (37):

(37)

Stem: sentential			Ending
Stem: lexical		Light verb stem	Inflexional suffix
Radical	Thematic suffix		
<i>taka-</i>	<i>-si</i>	—	<i>-0</i> [–PAST]
<i>nob-</i>	<i>-u</i>	—	<i>-0</i> [–PAST]
<i>taka-</i>	<i>-ki</i>	—	<i>-0</i> [–PAST]
<i>nob-</i>	<i>-u</i>	—	<i>-ru</i> [–PAST]
<i>taka-</i>	<i>-ku</i>	—	# [0PAST]
<i>nob-</i>	<i>-i</i>	—	# [0PAST]
<i>taka-</i>	<i>-ke</i>	—	<i>-reba</i>
<i>nob-</i>	<i>-u</i>	—	<i>-reba</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+ari-</i>	<i>-ki</i> [+PAST]
<i>nob-</i>	<i>-i</i>	—	<i>-ki</i> [+PAST]

As the VA thematic suffix shows allomorphy *-si* $\approx$ *-ki* $\approx$ *-ku* $\approx$ *-ke* depending on the syntactic environment, the V thematic suffix also exhibits the alternation *-i* $\approx$ *-u* according to the syntactic context in Antient language. Furthermore, the conclusive form *nob-u-0* of the classical *nobu*, corresponding to the modern *nobiru*, like the conclusive form *taka-si-0* of the classical *takasi*, corresponding to the modern *takai*, marked [–PAST] with *-0*, differently from modern Japanese.

The only instance where the parallelism in the suffixes breaks down is in the attributive forms *taka-ki-0* and *nob-u-ru*, but there seems to be a reason for this. In VA *takasi*, the opposition between the conclusive form and the attributive form was maintained by the alternation of the thematic suffix allomorphs *-si* $\approx$ *-ki*, as in *taka-si-0* and *taka-ki-0*, whereas in V *nobu*, the thematic

suffix allomorph was the same *-u* in both forms and the opposition was instead maintained by the alternation of the past tense suffix allomorphs *-o* $\approx$ *-ru*, as in *nob-u-o* and *nob-u-ru*.

7. Negative forms of verbal adjectives

Both verbs and verbal adjectives form their negative constructions by attaching the negator *nai* to their stems, as in *tabe-nai* "not to eat" from the verb *tabe-ru* and *takaku#nai* "not to be high/tall" from the verbal adjective *takai*. This negator belongs to the VA category, given its inflexion — namely, *na-i* $\approx$ *na-ku* $\approx$ *na-ku-te* $\approx$ *na-ku-temo* $\approx$ *na-ku-tatte* $\approx$ *na-k+at-ta* $\approx$ *na-k+at-tar-ō* $\approx$ *na-k+at-tar-a(ba)* $\approx$ *na-k+at-tar-i* $\approx$ *na-k+ar-ō* $\approx$ *na-ke-reba*, as in *tabe-na-i* $\approx$ *tabe-na-ku* $\approx$ *tabe-na-ku-te* $\approx$ *tabe-na-ku-temo* $\approx$ *tabe-na-ku-tatte* $\approx$ *tabe-na-k+at-ta* $\approx$ *tabe-na-k+at-tar-ō* $\approx$ *tabe-na-k+at-tar-a(ba)* $\approx$ *tabe-na-k+at-tar-i* $\approx$ *tabe-na-k+ar-ō* $\approx$ *tabe-na-ke-reba* and *takaku#na-i* $\approx$ *takaku#na-ku* $\approx$ *takaku#na-ku-te* $\approx$ *takaku#na-ku-temo* $\approx$ *takaku#na-ku-tatte* $\approx$ *takaku#na-k+at-ta* $\approx$ *takaku#na-k+at-tar-ō* $\approx$ *takaku#na-k+at-tar-a(ba)* $\approx$ *takaku#na-k+at-tar-i* $\approx$ *takaku#na-k+ar-ō* $\approx$ *takaku#na-ke-reba* — which is completely identical to that of the verbal adjective *takai* — namely, *taka-i* $\approx$ *taka-ku* $\approx$ *taka-ku-te* $\approx$ *taka-ku-temo* $\approx$ *taka-ku-tatte* $\approx$ *taka-k+at-ta* $\approx$ *taka-k+at-tar-ō* $\approx$ *taka-k+at-tar-a(ba)* $\approx$ *taka-k+at-tar-i* $\approx$ *taka-k+ar-ō* $\approx$ *taka-ke-reba*.

Although the above negative forms *tabe-nai* and *takaku#nai* may at first sight appear to exhibit the same morphosyntactic structure in that both consist of the stem + the negator *nai*, their structures in fact differ: **the negator *-na-i* in *tabe-nai* functions as a sequence of suffixes, whereas *na-i* in *takaku#nai* constitutes an autonomous word**, as illustrated in (38):

(38) <i>tabe-nai</i>	$\rightarrow$ <i>tabe=wa#<u>si-nai</u></i>	vs. <i>*tabe=wa#<u>nai</u></i>	"eat, it does not"
<i>tabe-nai</i>	$\rightarrow$ <i>tabe=mo#<u>si-nai</u></i>	vs. <i>*tabe=mo#<u>nai</u></i>	"it does not even eat"
<i>takaku#nai</i>	$\rightarrow$ <i>takaku=wa#<u>nai</u></i>	vs. <i>*takaku=wa#<u>ara-nai</u></i>	"high/tall, it is not"
<i>takaku#nai</i>	$\rightarrow$ <i>takaku=mo#<u>nai</u></i>	vs. <i>*takaku=mo#<u>ara-nai</u></i>	"it is not high/tall either"
cf. <i>takaku-te</i>	$\rightarrow$ <i>takaku=mo#<u>at-te</u></i>	vs. <i>*takaku=mo#<u>te</u></i>	in (15)

Both in *tabe-nai* and in *takaku#nai*, the enclitic particles *=mo* and *=wa* can be syntactically inserted immediately after the verbal stem *tabe-* and the verbal-adjective stem *takaku-*, just like the examples in (15) and (16) above. In the former case, the stem allomorph *si-* of the light verb *su-ru* must be inserted before the negator *-nai* as in *tabe=wa#si-nai* and *tabe=mo#si-nai*. The reason is as follows: since *-nai* function as a sequence of suffixes with respect to the stem *tabe-* in *tabe-nai*, once the particle *=mo* or *=wa* is inserted between the stem *tabe-* and the suffixes *-nai*,

the latter are left stemless. Consequently, an alternative formal stem must be supplied — namely, *si-* from *su-ru* — because the particles *=mo* and *=wa* cannot function as verbal stems for *-nai*.

In the latter case, by contrast, the same particle-insertion does not entail the insertion of the stem allomorph *ara-* from the light verb *ar-u* before *nai*, as in *takaku=wa#nai* and *takaku=mo#nai* — the allomorph expected in this morphosyntactic context for consonant-stem verbs such as *ar-u*, just as in *noma-nai* from the consonant-stem verb *nom-u* "to drink". This indicates that, in *takaku#nai*, the string *nai* functions as an autonomous word in relation to the other autonomous word *takaku* and therefore requires no verbal stem before it. The negator *nai* in *takaku#nai* is morphologically distinct, for example, from the suffix *-te* in *takaku-te*, which requires the stem allomorph *at-* from *ar-u* in the particle-inserted form *takaku=mo#at-te*, as already illustrated in (15).

The above reasoning is based on the insightful observation in Miyaoka (2002: 85, translated from Japanese and noted by the author): "whereas in *kaka-nai* 'it does not write' the suffixality of *-nai* is evident, since *si* (< *suru* 'to do') must be inserted as in *kaki=mo=si-nai* 'it does not even write', in *takaku=nai* the possibility of inserting the clitic *=mo* directly shows the cliticity of *=nai*."

The question arises, then, why the negator *nai* functions as a string of suffixes in *tabe-nai* but as an autonomous word in *takaku#nai*. In other words, why is there a lack of parallelism in morphological structure between the negative forms of verbs and those of verbal adjectives? This seems to be related to the fact that, in contemporary Japanese, the negative form of the verb *ar-u* is not the theoretically expected **ara-nai* — composed of the stem allomorph *ara-* from *ar-u* and the suffixes *-nai*, as in *noma-nai* from *nom-u* and *tabe-nai* from *tabe-ru* — but rather simply *nai*, which is not categorially a verb but a verbal adjective, as *takai* is. In other words, **the VA *nai* is a kind of suppletive form corresponding to the hypothetical negative form **ara-nai* within the inflexional or derivative paradigm of the verb *ar-u*.**

Building on this suppletion, let us hypothesise that, in the negation of verbal adjectives such as *takai*, it is likewise the suffixes *-nai* used for verbs — rather than the autonomous word *nai* — that are employed *underlyingly*, just as in the case of verbal negation. The derivation from *takai* + *-nai* to the attested surface form *takaku#nai* is then assumed to proceed as shown in (39):

(39) *takai* + *-nai* → **takaku#ara-nai* → *takaku#nai* (suppletion)

Since the verbal-adjective stem *takaku-* is not compatible with the negative suffix *-na*, which is verbal and subcategorises for purely verbal stems, the stem allomorph *ara-* from the light verb *ar-u* is inserted immediately after *takaku-*, thereby constituting, together with it, a composite

verbal stem *takaku#ara-* at the syntactic-derivation level. This composite stem as a whole acquires verbal status, due to the head-finality of this language, and can combine directly with the negative suffix *-na*, yielding the hypothetical intermediate string **takaku#ara-nai*, by the addition of the VA thematic suffix *-i* following *-na*. The verbal negative form **ara-nai* in **takaku#ara-nai* is, however, replaced by the VA *nai* at the surface level, as in *takaku#nai*, owing to the above obligatory suppletion within the inflexional or derivative paradigm of *ar-u*. On the other hand, since the word boundary *#* preceding **ara-nai* has remained intact, *takaku* and *nai* constitute two autonomous words.

In Old Japanese, on the other hand, the negative form of the *light* verb *ari*, which corresponds to the modern *ar-u*, was not *nasi*, which corresponds to the modern *nai*, but rather *ara-zu* with the negative suffix *-zu*, which corresponds to the modern *-nai* — just as the negative forms of *tabu-0*, corresponding to the modern *tabe-ru*, and *nom-u* were *tabe-zu* and *noma-zu*, respectively. Consequently, in the derivation of the negative form of *takasi*, which corresponds to the modern *takai*, the negative form *ara-zu* was not replaced by *nasi*, as shown in (40), where " $\rightarrow$ " indicates a failed derivation:

$$(40) \text{takaku} + \text{-zu} \rightarrow \text{takaku\#ara-zu} \rightarrow \text{takak+ara-zu} \text{ (no suppletion)} \\ \rightarrow * \text{takaku\#nasi}$$

The classical form *takaku#ara-zu* in (40) — from which the more synthetic form *takak+ara-zu* originated — is the structural equivalent of the modern hypothetical **takaku#ara-nai* in (39), whereas the classical hypothetical **takaku#nasi* in (40) is the structural equivalent of the modern attested *takaku#nai* in (39). Analysed in this way, the difference between Modern and Old Japanese in the formation of the negative forms of verbal adjectives lies in the modern suppletion of the negative form **ara-nai* of the light verb *ar-u* by *nai*, and the derivation schematised in (39) seems entirely plausible. Thus, the verbal-adjective negative form *takaku#nai-0*, with the endig *-0* specified as [–past] is assumed to be derived from the underlying form in (41):

$$(41) *[[[\text{takaku\#ara}]\text{nai}]\text{0}]$$

which exhibits the same structure as the verbal negative form in (42):

$$(42) [[[\text{tabe}]\text{nai}]\text{0}]$$

in that both strings are composed of [[[stem]suffixal negator]ending], apart from the difference in the degree of complexity between the composite stem **takaku#ara-* and the simplex *tabe-*.

Based on the above discussion, the *surface* negative forms of the VA *takai* can be analysed as in (43), where R, TS and S represent the radical, the thematic suffix and the stem, respectively:

(43)

Word		Word			
VA S: sentential		Negator S: sentential			Ending
VA S: lexical		Negator S: lexical		Light verb stem	Inflexional suffix(es)
VA R	VA TS	Negator R	Negator TS		
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-i</i>	—	<i>-0 [-PAST]</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-(k)i</i>	—	<i>-0 [-PAST]</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-ke</i>	—	<i>-reba</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-ku</i>	—	<i># [0 PAST]</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-ku</i>	—	<i>-te</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-ku</i>	—	<i>-temo</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-ku</i>	—	<i>-tatte</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-ta [+PAST]</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-ō</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-a(ba)</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-i</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+ar-</i>	<i>-ō</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+ar-</i>	<i>-e</i>

8. Syntactic and logical-propositional differences between *ku*-adjectives and *ni*-adjectives

The Japanese language has another type of adjectives — **nominal adjectives (NAs)**, also referred to as ***na*-adjectives** due to their attributive forms ending in *-na* (so-called *rentaikei* in Japanese linguistics), as in *sizuka-na heya* 'a quiet room'. According to Makino (2025), the core of NAs lies in the **NA stem in *-ni***, as in *sizuka-ni* 'to be quiet/silent' and *suki-ni* 'to like'. The element *-ni* is the NA thematic suffix, which functions as an **indispensable morphological and semantic constituent in the formation of the NA stem**. Thus, it seems **more appropriate to call NAs *ni*-adjectives** rather than *na*-adjectives, in accordance with this defining constituent *-ni*.

NA stems are lexical predicates that require arguments — the argument structure [theme] inherent in *sizuka-ni* and [experiencer, theme] in *suki-ni* — just as VA stems are, such as the one-argument predicate *tanosi-ku* 'to be enjoyable' with [theme] and the two-argument predicate *hosi-ku* 'to want' with [experiencer, theme]. However, outside small clauses functioning as complements of cognition or perception verbs, as in (44) and (45):

(44) *Watasi=wa [sore=o yukai-ni] omot-ta / kanzi-ta*. 'I found / felt [it pleasant].'

(45) *Watasi=wa [sore=o tanosi-ku] omot-ta / kanzi-ta*. 'I found / felt [it enjoyable].'

such NA stems cannot function as predicates in their bare *-ni* form, unlike VA stems, as shown in (46) – (49):

- (46) *Sore=wa tanosi-ku, yukai-de aru.* 'It is enjoyable, is pleasant.'
 (47) **Sore=wa tanosi-ku, yukai-ni.* 'It is enjoyable, pleasant.'
 (48) *Sore=wa yukai-de ari, tanosi-i.* 'It is pleasant, is enjoyable.'
 (49) **Sore-wa yukai-ni, tanosi-i.* 'It pleasant, is enjoyable.'

Consequently, an NA stem forms a nominal-adjective phrase containing PRO, as in [*PRO yukai-ni*], which serves as the complement of the adjunctive conjunction *-te*, yielding a ConjP [*PRO yukai-ni-te*]. This ConjP is then adjoined to the VP headed by the existential-verb stem *ar-* within a one-argument obligatory construction, as in [*sore_i=wa t_i ar-u*] 'it exists'. Only in this configuration can the NA stem function as the predicate of an independent sentence, as in [*sore_i=wa [PRO_i yukai-ni-te] t_i ar-u*], which surfaces as [*sore_i=wa [PRO_i yukai-de] t_i ar-u*] 'it is pleasant' (literally, 'it exists, being pleasant').

This construction is schematically representable as $\text{EXIST}(\text{sore}) \wedge \text{PLEASANT}(\text{sore})$, whose default negation pattern is partial, namely $\text{EXIST}(\text{sore}) \wedge \neg \text{PLEASANT}(\text{sore})$, with the scope of negation marked by the particle *=wa*, as indicated in (50) and (51):

- (50) *Sore=wa yukai-de aru:* $\text{EXIST}(\text{sore}) \wedge \text{PLEASANT}(\text{sore})$
 (51) *Sore=wa yukai-de=wa nai:* $\text{EXIST}(\text{sore}) \wedge \neg \text{PLEASANT}(\text{sore})$

The negative *-nai* therefore does not negate the entire construction *sore=wa yukai-de ar-u*, but instead targets only the adjunct *yukai-de*. Semantically, the sentence continues to assert the existence of *sore*, while negating only the claim that *sore* is *yukai-ni*. This follows from the adjunctive nature of *yukai-de*, which logically constitutes an additional proposition. The total negation in (52) is perceived as pragmatically infelicitous:

- (52) *Sore=wa yukai-de nai:* $\neg(\text{EXIST}(\text{sore}) \wedge \text{PLEASANT}(\text{sore}))$

By contrast, the default negation pattern of the VA construction in (53) is the total negation in (54), not the partial negation in (55):

- (53) *Sore=wa tanosi-i:* $\text{ENJOYABLE}(\text{sore})$
 (54) *Sore=wa tanosi-ku nai:* $\neg \text{ENJOYABLE}(\text{sore})$

(55) *Sore=wa tanosi-ku=wa nai*: $\neg \text{ENJOYABLE}(\text{sore}) \wedge \text{SOME-OTHER-STATE}(\text{sore})$

Accordingly, the VA construction *sore=wa tanosi-i* in (53) is fundamentally different from the NA construction *sore=wa yukai-de aru* in (50), in that the former is schematically representable as the atomic proposition **ENJOYABLE(sore)** — whose default negation pattern total, namely, $\neg \text{ENJOYABLE}(\text{sore})$ — whereas the latter can be represented as the conjunctive proposition **EXIST(sore) $\wedge$ PLEASANT(sore)** — whose default negation pattern partial, namely, **EXIST(sore) $\wedge$ $\neg$ PLEASANT(sore)**.

It follows from the above discussion that the stem allomorph *at-* of the verb *ar-u* occurring in the past form *tanosi-k-at-ta*, for example, does not instantiate the one-argument existential verb *ar-* 'to exist' used in nominal-adjectival sentences. Rather, it is an allomorph of the light verb stem *ar-* inserted as a matter of formal necessity in order to concatenate the verbal past-tense suffix *-ta* with the non-verbal VA stem *tanosi-ku*, as in *tanosi-ku + -ta* $\rightarrow$ *tanosi-ku + ar- + -ta* $\rightarrow$ *tanosi-k-at-ta*, as already discussed in Section 3. Thus, the syntactic derivation of the verbal-adjectival and nominal-adjectival constructions in (56) – (58) can be represented as follows:

(56) *Sore=wa tanosi-i*. 'It is enjoyable.'

Core construction

[*tanosi-ku*]
 $\rightarrow$ [*sore tanosi-ku*]
 $\rightarrow$ [[*sore tanosi-ku*] $T^0\emptyset$]
 $\rightarrow$ [[[*sore tanosi-ku*] $T^0\emptyset$] $Top^0\emptyset$]
 $\rightarrow$ [*sore_i=wa* [[[*t_i tanosi-ku*] $T^0\emptyset$] $Top^0\emptyset$]]
 $\rightarrow$ *sore=wa tanosi-i* (allomorphy or amalgam)

$T^0\emptyset$ is an alternative notation for the phonologically null inflexional suffix *-0* specified as [–PAST], already used in this paper. On the other hand, the category Topic, abbreviated as Top, is "a kind of functional category" that "can be considered to embed the IP into discourse and context" (Hasegawa 1999: 91, translated from Japanese), and is assumed in this paper to be realised by the phonologically null morpheme $Top^0\emptyset$, realised as *-0*. The element that moves to the specifier of $TopP$ is assigned the particle *=wa* and functions to indicate the topic of the sentence.

In the non-past in (56), the sequence [*-ku + $T^0\emptyset$*] is morphophonologically realised as *-i*, yielding the surface form *tanosi-i* from the same underlying predicational structure *tanosi-ku* when

combined with the non-past $T^0\emptyset$. The exponent *-i* may therefore be analysed either as a contextual allomorph *-i* of the VA thematic suffix *-ku* in the non-past environment or, alternatively, as an amalgamated realisation of the sequence $[-ku + T^0\emptyset]$, as stated in Section 5.

(57) *Sore=wa tanosi-k-at-ta*. 'It was enjoyable.'

Core construction

[*tanosi-ku*]
 → [*sore tanosi-ku*]
 → [[*sore tanosi-ku*] **-ta**]
 → [[*sore tanosi-ku*] **ar-** *-ta*] (light-verb-stem insertion)
 → [[[*sore tanosi-ku*] **ar-** *-ta*] **Top⁰∅**]
 → [*sore_i=wa* [[[*t_i tanosi-ku*] **ar-** *-ta*] **Top⁰∅**]]
 → *sore=wa tanosi-k-at-ta*

In the analysis of the VA past form in (57), the choice between the non-past $T^0\emptyset$, realised as *-0*, and the past suffix *-ta* is treated as a paradigmatic selection made at a single derivational stage, since the $[-\text{past}]$ *-0* and the $[\text{+past}]$ *-ta* constitute a paradigm whose terms acquire linguistic value solely through their opposition at the same point of speech. When the past suffix *-ta* is selected, it proves incompatible with non-verbal predicates such as *tanosi-ku*, thereby triggering the insertion of the light verb stem *ar-* as a repair strategy.

(58) *Sore=wa yukai-de aru*. '(lit.) It exists, being pleasant.'

Adjunct

Core construction

[<i>yukai-ni</i>]	[<i>ar-</i>]
→ [<i>PRO yukai-ni</i>]	→ [<i>sore ar-</i>]
→ [[<i>PRO yukai-ni</i>] -te]	
	→ [[[<i>PRO_i yukai-ni</i>] -te] [<i>sore_i ar-</i>]] (adjunct embedding)
	→ [[[[<i>PRO_i yukai-ni</i>] -te] [<i>sore_i ar-</i>]] -ru]
	→ [[[[[[<i>PRO_i yukai-ni</i>] -te] [<i>sore_i ar-</i>]] -ru] Top⁰∅]
	→ [<i>sore_i=wa</i> [[[[[[<i>PRO_i yukai-ni</i>] -te] [<i>t_i ar-</i>]] -ru] Top⁰∅]]
	→ <i>sore=wa yukai-de ar-u</i>

9. Analysis summary — Conclusion

(i) The morphosyntactic structure of the inflected forms of verbal adjectives is as shown in (59), where 'lit.' and 'coll.' abbreviate 'literal form' and 'colloquial form', respectively:

(59)

Underlying form				Surface form
Compound stem: sentential			Ending	
VA stem: lexical		Light verb stem	Inflex. suff.	
VA rad.	VA them. suff.			
<i>taka-</i>	<i>-ku</i>	—	# [0 PAST]	<i>taka-ku</i>
<i>taka-</i>	<i>-ku</i>	—	-0 [-PAST]	<i>taka-i</i>
<i>taka-</i>	<i>-ku</i>	—	-0 [-PAST]	<i>taka-ki</i> (lit.) <i>taka-i</i>
<i>taka-</i>	<i>-ku</i>	—	- <i>reba</i>	<i>taka-ke-reba</i>
<i>taka-</i>	<i>-ku</i>	—	- <i>te</i>	<i>taka-ku-te</i> <i>taka-ku-tte</i> (coll.)
<i>taka-</i>	<i>-ku</i>	—	- <i>temo</i>	<i>taka-ku-temo</i> <i>taka-ku-ttemo</i> (coll.)
<i>taka-</i>	<i>-ku</i>	—	- <i>tatte</i>	<i>taka-ku-tatte</i> <i>taka-ku-ttatte</i> (coll.)
<i>taka-</i>	<i>-ku</i>	<i>ar-</i>	- <i>tar</i> [+PAST]	<i>taka-k+at-ta</i>
<i>taka-</i>	<i>-ku</i>	<i>ar-</i>	- <i>tar-jō</i>	<i>taka-k+at-tar-ō</i>
<i>taka-</i>	<i>-ku</i>	<i>ar-</i>	- <i>tar-a(ba)</i>	<i>taka-k+at-tar-a(ba)</i>
<i>taka-</i>	<i>-ku</i>	<i>ar-</i>	- <i>tar-i</i>	<i>taka-k+at-tar-i</i>
<i>taka-</i>	<i>-ku</i>	<i>ar-</i>	- <i>jō</i>	<i>taka-k+ar-ō</i>
<i>taka-</i>	<i>-ku</i>	<i>ar-</i>	- <i>e</i>	<i>taka-k+ar-e</i> (lit.) <i>taka-ku#ar-e</i>
<i>taka-</i>	<i>-ku</i>	<i>ar-</i>	#- <i>tutu</i>	<i>taka-ku#ari-tutu</i>
<i>taka-</i>	<i>-ku</i>	<i>ar-</i>	#- <i>nagara</i>	<i>taka-ku#ari-nagara</i>

Unlike the previous schematic representations, the left-hand column lists the underlying forms, and the right-hand column lists the corresponding surface forms, so as to make explicit the allomorphy involved in the derivation. Note that the underlying representations of the inflexional suffixes *-ta* and *-ō* are analysed as *-tar* and *-jō*, respectively, and that the suffixes *-te*, *-temo*, *-tatte* may be realised as *-tte*, *-ttemo*, *-ttatte* in colloquial registers, with emphatic consonantal gemination.

(ii) The VA radical is the element that encodes the lexical meaning specific to each verbal adjective. It can appear independently as a constituent of derived or compound words — for example, *taka-* in *taka-sa* "height", *taka-mi* "loftiness", *taka+nozomi* "high hope" and *taka+dai* "high platform" — but it cannot, on its own, be used as a syntactic constituent belonging to the category VA at the sentential level — for example, **Kare =wa se ga taka* "He is tall"; **Asoko ni se no taka otoko ga iru* "There is a tall man over there"; **Koko kara taka sobieru tō ga mieru*

"From here, a tower rising high can be seen"; **Se =wa taka-reba ii to yū wake de =wa nai* "It is not always the case that it is good if the height is tall".

(iii) The function of the VA thematic suffix $-i \approx -(k)i \approx -ku \approx -ke$ is to licence the VA radical as a syntactic constituent of category VA within sentential phrase structure. As noted in (ii), *taka-* on its own cannot be used as a VA in a sentence. Only when it combines with $-i$, $-(k)i$, $-ku$ or $-ke$ — forming *taka-i*, *taka-(k)i*, *taka-ku* or *taka-ke*, for example — does it become syntactically eligible to function as a constituent of category VA — e.g. *Kare =wa se ga takai*; *Asoko ni se no takai otoko ga iru*; *Koko kara takaku sobieru tō ga mieru*; *Se =wa takake-reba ii to yū wake de =wa nai*.

(iv) The form in which the VA thematic suffix appears — $-i$, $-(k)i$, $-ku$ or $-ke$ — is determined by the syntactic environment in which the VA stem occurs, as shown in (60) – (63):

(60) When a VAP headed by a VA stem serves as the complement of a TP headed by the inflexional suffix *-reba*: $-ku \rightarrow -ke$

(61) When a VAP headed by a VA stem serves as the complement of a TP headed by -0 [–PAST], and that TP is embedded within an NP: $-ku \rightarrow -(k)i$

(62) When a VAP headed by a VA stem serves as the complement of a TP headed by -0 [–PAST], and that TP is not embedded within an NP: $-ku \rightarrow -i$

(63) In all other cases: $-ku$

(v) At the surface level, $-ku$ exhibits the phonological alternation $[-ku] \approx [-k]$, with vowel deletion occurring only when the VA stem combines with the stem *ar-* and its allomorphs of the light verb *ar-u*, resulting in the form $[-k]$. This vowel deletion can be taken to indicate that the VA stem in $-ku$ and the light verb stem *ar-* have come to form a single compound stem at the sentential level.

(vi) It is also possible to interpret $-i$ and $-(k)i$ as a synchronic amalgam of the VA thematic suffix and the tense suffix [–PAST]. Under this analysis, $-i$ and $-(k)i$ are not allomorphs exclusively of the VA thematic suffix but simultaneously encode [–PAST], thereby partially overlapping with the traditional analysis.

(vii) As shown in Section 5, there are significant structural parallels between the inflected forms of verbal adjectives — e.g. *takai* and its classical form *takasi* — and those of verbs, especially vowel-stem verbs — e.g. *nobiru* and its classical form *nobu*. Here, the vowel-stem verb *kuru* "to

come" is further examined, and the morphosyntactic structure of its inflected forms, semantically corresponding to those of the verbal adjective *takai* exemplified in (59), is shown in (64). Again, a strong parallelism in morphosyntactic structure can be observed:

(64)

Underlying form			Surface form
V stem		Ending	
V radical	V thematic suffix	Inflexional suffix(es)	
<i>k-</i>	<i>-i</i>	<i>-ru</i> [-PAST]	<i>k-u-ru</i>
<i>k-</i>	<i>-i</i>	<i>-reba</i>	<i>k-u-reba</i>
<i>k-</i>	<i>-i</i>	# [0 PAST]	<i>k-i</i>
<i>k-</i>	<i>-i</i>	<i>-te</i>	<i>k-i-te</i>
<i>k-</i>	<i>-i</i>	<i>-temo</i>	<i>k-i-temo</i>
<i>k-</i>	<i>-i</i>	<i>-tatte</i>	<i>k-i-tatte</i>
<i>k-</i>	<i>-i</i>	<i>-tar</i> [+PAST]	<i>k-i-ta</i>
<i>k-</i>	<i>-i</i>	<i>-tar-ō</i>	<i>k-i-tar-ō</i>
<i>k-</i>	<i>-i</i>	<i>-tar-a(ba)</i>	<i>k-i-tar-a(ba)</i>
<i>k-</i>	<i>-i</i>	<i>-tar-i</i>	<i>k-i-tar-i</i>
<i>k-</i>	<i>-i</i>	<i>-jō</i>	<i>k-o-jō</i>
<i>k-</i>	<i>-i</i>	<i>-jo</i>	<i>k-o-i</i>
<i>k-</i>	<i>-i</i>	#-tutu	<i>k-i-tutu</i>
<i>k-</i>	<i>-i</i>	#-nagara	<i>k-i-nagara</i>

In vowel-stem verbs, the stem-final vowels — *-u* $\approx$ *-i* $\approx$ *-o* in the inflected forms of the verb [[[*k*]*u*]*ru*] as well as *-i* in those of the verb [[[*nob*]*i*]*ru*] in (34) and *-u* $\approx$ *-i* in those of the corresponding classical verb [[[*nob*]*u*]*0*] in (35) — can be understood as functioning as the thematic suffix. For further details on the morphosyntactic structure of verbs, see the Appendix.

(viii) As discussed in Section 6, in the surface negative forms of verbal adjectives such as *takaku#nai-0*, *takaku#nak+at-ta*, *takaku#nak+ar-ō*, *takaku#nake-reba*, the constituents which are morphologically autonomous words such as *nai-0*, *nak+at-ta*, *nak+ar-ō*, *nake-reba* are assumed to be derived from the underlying strings such as **ara-nai-0*, **ara-nak+at-ta*, **ara-nak+ar-ō*, **ara-nake-reba*, due to the obligatory suppletion of the negative forms of the light verb *ar-u* by the inflected forms of the VA *nai*. Thus, the hypothetical underlying forms corresponding to the above surface forms are **[[[takaku#ara]nai]0]*, **[[[takaku#ara]nak+at]ta]*, **[[[takaku#ara]nak+ar]ō]*, **[[[takaku#ara]nake]reba]*, which exhibit the same structure — that is, [[[stem]suffixal negator]ending] — as the verbal negative forms such as [[[*tabe*]nai]0], [[[*tabe*]nak+at]ta], [[[*tabe*]nak+ar]ō], [[[*tabe*]nake]reba], apart from the difference in the degree of structural complexity between the stems **[takaku#ara-]* and *[tabe-]*.

(ix) Whereas the she nominal-adjectival sentences consist syntactically of an existence-asserting, one-argument obligatory construction [$NP_i=wa\ t_i\ ar-TENSE$] combined with a state-specifying adjunctive ConjP [$PRO_i\ NA\ radical-ni-te$] and can be schematically represented as conjunctive $EXIST(NP) \wedge NA-STATE(NP)$, whose default negation pattern is partial, namely $EXIST(NP) \wedge \neg NA-STATE(NP)$, the verbal-adjectival sentences consist only of a state-specifying obligatory construction [$NP_i=wa\ t_i\ VA\ radical-ku-TENSE$] without any adjunct and can be represented as atomic $VA-STATE(NP)$, whose default negation pattern is total, namely $\neg VA-STATE(NP)$.

Appendix — Computation of Japanese inflected verbal forms from underlying representations

This appendix presents an ordered computation of inflected verbal forms from their underlying representations. Its purpose is not to provide an exhaustive account of all verbal morphology in Japanese, but to make it explicit that a substantial set of surface forms can be derived consistently from underlying forms by means of a small number of ordered operations. In this respect, the appendix is intended as a supplementary demonstration of the structural parallelism between verbal adjectives and true verbs argued for in the main text. The surface forms are derived from the underlying representations by the ordered application of the following operations.

Ordered operations

1. Insert **-ru** between **-Ø** and **-mai**.

Optionally, insert **-ru** between **-i** and **-mai** in the **s-irregular** and **k-irregular** verbs.

2. Before **-na***, **-zu**, change **Ø** to **a**.

Before **-ru**, **-reba**, change **i** to **u** in the **s-irregular** and **k-irregular** verbs.

Before **-jo**, **-zu**, **-sas***, **-rar***, change **i** to **e** in the **s-irregular** verb.

Before **-jō**, **-jo**, **-na***, **-zu**, **-sas***, **-rar***, **-mai**, change **i** to **o** in the **k-irregular** verb.

Optionally, before **-mai**, change **i** to **u** in the **s-irregular** verb.

Optionally, before **-ru** and **-reba**, change **e** to **u** in the verbs **Ø-e-ru** 'gain' and **p-e-ru** [heru] 'pass (time)'.

3. Delete all remaining instances of **-Ø**.
4. Delete suffix-initial **s**, **r**, **j** after any consonant.

5. In *-te*, *-temo*, *-tatte*, *-tar**, change suffix-initial *t* to *d* after **b*, **g*, **m*, **n*.

6. Before *-de*, *-demo*, *-datte*, *-dar**, change *b*, *m* to *n*.

Before *-de*, *-demo*, *-datte*, *-dar**, change *g* to *i*.

7. Before *-te*, *-temo*, *-tatte*, *-tar**, change *ow* to *ō*.

8. Before *-te*, *-temo*, *-tatte*, *-tar**, change *r*, *w* to *t*.

Before *-te*, *-temo*, *-tatte*, *-tar**, change *s* to *si*.

Before *-te*, *-temo*, *-tatte*, *-tar**, change *k* to *i*, except in the verb *ik-u* 'go', underlyingly *ik-Ø-ru*, where *k* changes to *t*.

9. Insert *i* immediately after a consonant before #.

10. Delete all *w* except before *a**.

11. Delete word-final *r*.

12. Change *-jo* to *-i* in the *k-irregular* verb.

Ordering: 1 → 2 → 3 → {(5 → 6), (7 → 8)} in either order → 9 → {10, 11, 12} in any order.

In the schematic representation below, consonant-stem verbs have a verbal stem whose underlying V thematic suffix is *-Ø*, whereas vowel-stem verbs have a verbal stem whose underlying V thematic suffix is *-i* or *-e*.

Underlying form			Surface form
V stem		Ending	
V radical	V thematic suffix	Inflexional or derivational suffix(es)	
<i>k-</i>	<i>-i</i>	<i>-ru</i> [−PAST]	<i>k-i-ru</i> 'wear'
<i>k-</i>	<i>-i</i>	<i>-reba</i>	<i>k-i-reba</i>
<i>k-</i>	<i>-i</i>	# [0 PAST]	<i>k-i</i>
<i>k-</i>	<i>-i</i>	<i>#-tutu</i>	<i>k-i-tutu</i>
<i>k-</i>	<i>-i</i>	<i>#-nagara</i>	<i>k-i-nagara</i>
<i>k-</i>	<i>-i</i>	<i>#-gatera</i>	<i>k-i-gatera</i>
<i>k-</i>	<i>-i</i>	<i>#-ta-ku</i>	<i>k-i-ta-ku</i>
<i>k-</i>	<i>-i</i>	<i>-te</i>	<i>k-i-te</i>
<i>k-</i>	<i>-i</i>	<i>-temo</i>	<i>k-i-temo</i>
<i>k-</i>	<i>-i</i>	<i>-tatte</i>	<i>k-i-tatte</i>
<i>k-</i>	<i>-i</i>	<i>-tar</i> [+PAST]	<i>k-i-ta</i>
<i>k-</i>	<i>-i</i>	<i>-tar-jō</i>	<i>k-i-tar-ō</i>
<i>k-</i>	<i>-i</i>	<i>-tar-a(ba)</i>	<i>k-i-tar-a(ba)</i>
<i>k-</i>	<i>-i</i>	<i>-tar-i</i>	<i>k-i-tar-i</i>

<i>k-</i>	<i>-i</i>	<i>-jō</i>
<i>k-</i>	<i>-i</i>	<i>-ro</i>
<i>k-</i>	<i>-i</i>	<i>-jo</i>
<i>k-</i>	<i>-i</i>	<i>-na-ku</i>
<i>k-</i>	<i>-i</i>	<i>-zu</i>
<i>k-</i>	<i>-i</i>	<i>-sas-e</i>
<i>k-</i>	<i>-i</i>	<i>-rar-e</i>
<i>k-</i>	<i>-i</i>	<i>-mai</i>

<i>k-i-jō</i>
<i>k-i-ro</i>
<i>k-i-jo</i> (lit.)
<i>k-i-na-ku</i>
<i>k-i-zu</i> (lit.)
<i>k-i-sas-e</i>
<i>k-i-rar-e</i>
<i>k-i-mai</i>

<i>tab-</i>	<i>-e</i>	<i>-ru</i> [−PAST]
<i>tab-</i>	<i>-e</i>	<i>-reba</i>
<i>tab-</i>	<i>-e</i>	# [0 PAST]
<i>tab-</i>	<i>-e</i>	#- <i>tutu</i>
<i>tab-</i>	<i>-e</i>	#- <i>nagara</i>
<i>tab-</i>	<i>-e</i>	#- <i>gatera</i>
<i>tab-</i>	<i>-e</i>	#- <i>ta-ku</i>
<i>tab-</i>	<i>-e</i>	<i>-te</i>
<i>tab-</i>	<i>-e</i>	<i>-temo</i>
<i>tab-</i>	<i>-e</i>	<i>-tatte</i>
<i>tab-</i>	<i>-e</i>	<i>-tar</i> [+PAST]
<i>tab-</i>	<i>-e</i>	<i>-tar-jō</i>
<i>tab-</i>	<i>-e</i>	<i>-tar-a(ba)</i>
<i>tab-</i>	<i>-e</i>	<i>-tar-i</i>
<i>tab-</i>	<i>-e</i>	<i>-jō</i>
<i>tab-</i>	<i>-e</i>	<i>-ro</i>
<i>tab-</i>	<i>-e</i>	<i>-jo</i>
<i>tab-</i>	<i>-e</i>	<i>-na-ku</i>
<i>tab-</i>	<i>-e</i>	<i>-zu</i>
<i>tab-</i>	<i>-e</i>	<i>-sas-e</i>
<i>tab-</i>	<i>-e</i>	<i>-rar-e</i>
<i>tab-</i>	<i>-e</i>	<i>-mai</i>

<i>tab-e-ru</i> 'eat'
<i>tab-e-reba</i>
<i>tab-e</i>
<i>tab-e-tutu</i>
<i>tab-e-nagara</i>
<i>tab-e-gatera</i>
<i>tab-e-ta-ku</i>
<i>tab-e-te</i>
<i>tab-e-temo</i>
<i>tab-e-tatte</i>
<i>tab-e-ta</i>
<i>tab-e-tar-ō</i>
<i>tab-e-tar-a(ba)</i>
<i>tab-e-tar-i</i>
<i>tab-e-jō</i>
<i>tab-e-ro</i>
<i>tab-e-jo</i> (lit.)
<i>tab-e-na-ku</i>
<i>tab-e-zu</i> (lit.)
<i>tab-e-sas-e</i>
<i>tab-e-rar-e</i>
<i>tab-e-mai</i>

<i>Ø-</i>	<i>-e</i>	<i>-ru</i> [−PAST]
<i>Ø-</i>	<i>-e</i>	<i>-reba</i>
<i>Ø-</i>	<i>-e</i>	# [0 PAST]
<i>Ø-</i>	<i>-e</i>	#- <i>tutu</i>
<i>Ø-</i>	<i>-e</i>	#- <i>nagara</i>
<i>Ø-</i>	<i>-e</i>	#- <i>gatera</i>
<i>Ø-</i>	<i>-e</i>	#- <i>ta-ku</i>
<i>Ø-</i>	<i>-e</i>	<i>-te</i>
<i>Ø-</i>	<i>-e</i>	<i>-temo</i>
<i>Ø-</i>	<i>-e</i>	<i>-tatte</i>
<i>Ø-</i>	<i>-e</i>	<i>-tar</i> [+PAST]
<i>Ø-</i>	<i>-e</i>	<i>-tar-jō</i>
<i>Ø-</i>	<i>-e</i>	<i>-tar-a(ba)</i>
<i>Ø-</i>	<i>-e</i>	<i>-tar-i</i>
<i>Ø-</i>	<i>-e</i>	<i>-jō</i>
<i>Ø-</i>	<i>-e</i>	<i>-ro</i>

<i>e-ru</i> 'gain'
<i>u-ru</i> (lit.)
<i>e-reba</i>
<i>u-reba</i> (lit.)
<i>e</i>
<i>e-tutu</i>
<i>e-nagara</i>
<i>e-gatera</i>
<i>e-ta-ku</i>
<i>e-te</i>
<i>e-temo</i>
<i>e-tatte</i>
<i>e-ta</i>
<i>e-tar-ō</i>
<i>e-tar-a(ba)</i>
<i>e-tar-i</i>
<i>e-jō</i>
<i>e-ro</i>

<i>Ø-</i>	<i>-e</i>	<i>-jo</i>
<i>Ø-</i>	<i>-e</i>	<i>-na-ku</i>
<i>Ø-</i>	<i>-e</i>	<i>-zu</i>
<i>Ø-</i>	<i>-e</i>	<i>-sas-e</i>
<i>Ø-</i>	<i>-e</i>	<i>-rar-e</i>
<i>Ø-</i>	<i>-e</i>	<i>-mai</i>

<i>e-jo</i> (lit.)
<i>e-na-ku</i>
<i>e-zu</i> (lit.)
<i>e-sas-e</i>
<i>e-rar-e</i>
<i>e-mai</i>

<i>k-</i>	<i>-i</i>	<i>-ru</i> [–PAST]
<i>k-</i>	<i>-i</i>	<i>-reba</i>
<i>k-</i>	<i>-i</i>	# [0 PAST]
<i>k-</i>	<i>-i</i>	<i>#-tutu</i>
<i>k-</i>	<i>-i</i>	<i>#-nagara</i>
<i>k-</i>	<i>-i</i>	<i>#-gatera</i>
<i>k-</i>	<i>-i</i>	<i>#-ta-ku</i>
<i>k-</i>	<i>-i</i>	<i>-te</i>
<i>k-</i>	<i>-i</i>	<i>-temo</i>
<i>k-</i>	<i>-i</i>	<i>-tatte</i>
<i>k-</i>	<i>-i</i>	<i>-tar</i> [+PAST]
<i>k-</i>	<i>-i</i>	<i>-tar-jō</i>
<i>k-</i>	<i>-i</i>	<i>-tar-a(ba)</i>
<i>k-</i>	<i>-i</i>	<i>-tar-i</i>
<i>k-</i>	<i>-i</i>	<i>-jō</i>
<i>k-</i>	<i>-i</i>	<i>-jo</i>
<i>k-</i>	<i>-i</i>	<i>-na-ku</i>
<i>k-</i>	<i>-i</i>	<i>-zu</i>
<i>k-</i>	<i>-i</i>	<i>-sas-e</i>
<i>k-</i>	<i>-i</i>	<i>-rar-e</i>
<i>k-</i>	<i>-i</i>	<i>-mai</i>

<i>k-u-ru</i> 'come' (<i>k</i> -irregular)
<i>k-u-reba</i>
<i>k-i</i>
<i>k-i-tutu</i>
<i>k-i-nagara</i>
<i>k-i-gatera</i>
<i>k-i-ta-ku</i>
<i>k-i-te</i>
<i>k-i-temo</i>
<i>k-i-tatte</i>
<i>k-i-ta</i>
<i>k-i-tar-ō</i>
<i>k-i-tar-a(ba)</i>
<i>k-i-tar-i</i>
<i>k-o-jō</i>
<i>k-o-i</i>
<i>k-o-na-ku</i>
<i>k-o-zu</i>
<i>k-o-sas-e</i>
<i>k-o-rar-e</i>
<i>k-o-mai</i> (lit.)
<i>k-u-ru-mai</i>

<i>s-</i>	<i>-i</i>	<i>-ru</i> [–PAST]
<i>s-</i>	<i>-i</i>	<i>-reba</i>
<i>s-</i>	<i>-i</i>	# [0 PAST]
<i>s-</i>	<i>-i</i>	<i>#-tutu</i>
<i>s-</i>	<i>-i</i>	<i>#-nagara</i>
<i>s-</i>	<i>-i</i>	<i>#-gatera</i>
<i>s-</i>	<i>-i</i>	<i>#-ta-ku</i>
<i>s-</i>	<i>-i</i>	<i>-te</i>
<i>s-</i>	<i>-i</i>	<i>-temo</i>
<i>s-</i>	<i>-i</i>	<i>-tatte</i>
<i>s-</i>	<i>-i</i>	<i>-tar</i> [+PAST]
<i>s-</i>	<i>-i</i>	<i>-tar-jō</i>
<i>s-</i>	<i>-i</i>	<i>-tar-a(ba)</i>
<i>s-</i>	<i>-i</i>	<i>-tar-i</i>
<i>s-</i>	<i>-i</i>	<i>-jō</i>
<i>s-</i>	<i>-i</i>	<i>-ro</i>
<i>s-</i>	<i>-i</i>	<i>-jo</i>
<i>s-</i>	<i>-i</i>	<i>-mai</i>
<i>s-</i>	<i>-i</i>	<i>-na-ku</i>

<i>s-u-ru</i> 'do' (<i>s</i> -irregular)
<i>s-u-reba</i>
<i>s-i</i>
<i>s-i-tutu</i>
<i>s-i-nagara</i>
<i>s-i-gatera</i>
<i>s-i-ta-ku</i>
<i>s-i-te</i>
<i>s-i-temo</i>
<i>s-i-tatte</i>
<i>s-i-ta</i>
<i>s-i-tar-ō</i>
<i>s-i-tar-a(ba)</i>
<i>s-i-tar-i</i>
<i>s-i-jō</i>
<i>s-i-ro</i>
<i>s-e-jo</i> (lit.)
<i>s-i-mai</i> (lit.)
<i>s-u-mai</i> (lit.)
<i>s-u-ru-mai</i>
<i>s-i-na-ku</i>

<i>s-</i>	<i>-i</i>	<i>-zu</i>
<i>s-</i>	<i>-i</i>	<i>-sas-e</i>
<i>s-</i>	<i>-i</i>	<i>-rar-e</i>

<i>kes-</i>	<i>-Ø</i>	<i>-ru</i> [–PAST]
<i>kes-</i>	<i>-Ø</i>	<i>-reba</i>
<i>kes-</i>	<i>-Ø</i>	# [0 PAST]
<i>kes-</i>	<i>-Ø</i>	#-tutu
<i>kes-</i>	<i>-Ø</i>	#-nagara
<i>kes-</i>	<i>-Ø</i>	#-gatera
<i>kes-</i>	<i>-Ø</i>	#-ta-ku
<i>kes-</i>	<i>-Ø</i>	<i>-te</i>
<i>kes-</i>	<i>-Ø</i>	<i>-temo</i>
<i>kes-</i>	<i>-Ø</i>	<i>-tatte</i>
<i>kes-</i>	<i>-Ø</i>	<i>-tar</i> [+PAST]
<i>kes-</i>	<i>-Ø</i>	<i>-tar-jō</i>
<i>kes-</i>	<i>-Ø</i>	<i>-tar-a(ba)</i>
<i>kes-</i>	<i>-Ø</i>	<i>-tar-i</i>
<i>kes-</i>	<i>-Ø</i>	<i>-jō</i>
<i>kes-</i>	<i>-Ø</i>	<i>-e</i>
<i>kes-</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>kes-</i>	<i>-Ø</i>	<i>-zu</i>
<i>kes-</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>kes-</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>kes-</i>	<i>-Ø</i>	<i>-mai</i>

<i>kak-</i>	<i>-Ø</i>	<i>-ru</i> [–PAST]
<i>kak-</i>	<i>-Ø</i>	<i>-reba</i>
<i>kak-</i>	<i>-Ø</i>	# [0 PAST]
<i>kak-</i>	<i>-Ø</i>	#-tutu
<i>kak-</i>	<i>-Ø</i>	#-nagara
<i>kak-</i>	<i>-Ø</i>	#-gatera
<i>kak-</i>	<i>-Ø</i>	#-ta-ku
<i>kak-</i>	<i>-Ø</i>	<i>-te</i>
<i>kak-</i>	<i>-Ø</i>	<i>-temo</i>
<i>kak-</i>	<i>-Ø</i>	<i>-tatte</i>
<i>kak-</i>	<i>-Ø</i>	<i>-tar</i> [+PAST]
<i>kak-</i>	<i>-Ø</i>	<i>-tar-jō</i>
<i>kak-</i>	<i>-Ø</i>	<i>-tar-a(ba)</i>
<i>kak-</i>	<i>-Ø</i>	<i>-tar-i</i>
<i>kak-</i>	<i>-Ø</i>	<i>-jō</i>
<i>kak-</i>	<i>-Ø</i>	<i>-e</i>
<i>kak-</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>kak-</i>	<i>-Ø</i>	<i>-zu</i>
<i>kak-</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>kak-</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>kak-</i>	<i>-Ø</i>	<i>-mai</i>

<i>s-e-zu</i>
<i>s-e-sas-e</i>
<i>s-as-e</i>
<i>s-e-rar-e</i>
<i>s-ar-e</i>

<i>kes-u</i> 'erase, delete'
<i>kes-eba</i>
<i>kesi</i>
<i>kesi-tutu</i>
<i>kesi-nagara</i>
<i>kesi-gatera</i>
<i>kesi-ta-ku</i>
<i>kesi-te</i>
<i>kesi-temo</i>
<i>kesi-tatte</i>
<i>kesi-ta</i>
<i>kesi-tar-ō</i>
<i>kesi-tar-a(ba)</i>
<i>kesi-tar-i</i>
<i>kes-ō</i>
<i>kes-e</i>
<i>kes-a-na-ku</i>
<i>kes-a-zu</i> (lit.)
<i>kes-as-e</i>
<i>kes-ar-e</i>
<i>kes-u-mai</i>

<i>kak-u</i> 'write'
<i>kak-eba</i>
<i>kaki</i>
<i>kaki-tutu</i>
<i>kaki-nagara</i>
<i>kaki-gatera</i>
<i>kaki-ta-ku</i>
<i>kai-te</i>
<i>kai-temo</i>
<i>kai-tatte</i>
<i>kai-ta</i>
<i>kai-tar-ō</i>
<i>kai-tar-a(ba)</i>
<i>kai-tar-i</i>
<i>kak-ō</i>
<i>kak-e</i>
<i>kak-a-na-ku</i>
<i>kak-a-zu</i>
<i>kak-as-e</i>
<i>kak-ar-e</i>
<i>kak-u-mai</i>

<i>ik-</i>	<i>-Ø</i>	<i>-ru</i> [–PAST]
<i>ik-</i>	<i>-Ø</i>	<i>-reba</i>
<i>ik-</i>	<i>-Ø</i>	<i>#</i> [0 PAST]
<i>ik-</i>	<i>-Ø</i>	<i>#-tutu</i>
<i>ik-</i>	<i>-Ø</i>	<i>#-nagara</i>
<i>ik-</i>	<i>-Ø</i>	<i>#-gatera</i>
<i>ik-</i>	<i>-Ø</i>	<i>#-ta-ku</i>
<i>ik-</i>	<i>-Ø</i>	<i>-te</i>
<i>ik-</i>	<i>-Ø</i>	<i>-temo</i>
<i>ik-</i>	<i>-Ø</i>	<i>-tatte</i>
<i>ik-</i>	<i>-Ø</i>	<i>-tar</i> [+PAST]
<i>ik-</i>	<i>-Ø</i>	<i>-tar-jō</i>
<i>ik-</i>	<i>-Ø</i>	<i>-tar-a(ba)</i>
<i>ik-</i>	<i>-Ø</i>	<i>-tar-i</i>
<i>ik-</i>	<i>-Ø</i>	<i>-jō</i>
<i>ik-</i>	<i>-Ø</i>	<i>-e</i>
<i>ik-</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>ik-</i>	<i>-Ø</i>	<i>-zu</i>
<i>ik-</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>ik-</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>ik-</i>	<i>-Ø</i>	<i>-mai</i>

<i>ik-u</i> 'go'
<i>ik-eba</i>
<i>iki</i>
<i>iki-tutu</i>
<i>iki-nagara</i>
<i>iki-gatera</i>
<i>iki-ta-ku</i>
<i>it-te</i>
<i>it-temo</i>
<i>it-tatte</i>
<i>it-ta</i>
<i>it-tar-ō</i>
<i>it-tar-a(ba)</i>
<i>it-tar-i</i>
<i>ik-ō</i>
<i>ik-e</i>
<i>ik-a-na-ku</i>
<i>ik-a-zu</i>
<i>ik-as-e</i>
<i>ik-ar-e</i>
<i>ik-u-mai</i>

<i>nug-</i>	<i>-Ø</i>	<i>-ru</i> [–PAST]
<i>nug-</i>	<i>-Ø</i>	<i>-reba</i>
<i>nug-</i>	<i>-Ø</i>	<i>#</i> [0 PAST]
<i>nug-</i>	<i>-Ø</i>	<i>#-tutu</i>
<i>nug-</i>	<i>-Ø</i>	<i>#-nagara</i>
<i>nug-</i>	<i>-Ø</i>	<i>#-gatera</i>
<i>nug-</i>	<i>-Ø</i>	<i>#-ta-ku</i>
<i>nug-</i>	<i>-Ø</i>	<i>-te</i>
<i>nug-</i>	<i>-Ø</i>	<i>-temo</i>
<i>nug-</i>	<i>-Ø</i>	<i>-tatte</i>
<i>nug-</i>	<i>-Ø</i>	<i>-tar</i> [+PAST]
<i>nug-</i>	<i>-Ø</i>	<i>-tar-jō</i>
<i>nug-</i>	<i>-Ø</i>	<i>-tar-a(ba)</i>
<i>nug-</i>	<i>-Ø</i>	<i>-tar-i</i>
<i>nug-</i>	<i>-Ø</i>	<i>-jō</i>
<i>nug-</i>	<i>-Ø</i>	<i>-e</i>
<i>nug-</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>nug-</i>	<i>-Ø</i>	<i>-zu</i>
<i>nug-</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>nug-</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>nug-</i>	<i>-Ø</i>	<i>-mai</i>

<i>nug-u</i> 'take off (clothes)'
<i>nug-eba</i>
<i>nugi</i>
<i>nugi-tutu</i>
<i>nugi-nagara</i>
<i>nugi-gatera</i>
<i>nugi-ta-ku</i>
<i>nui-de</i>
<i>nui-demo</i>
<i>nui-datte</i>
<i>nui-da</i>
<i>nui-dar-ō</i>
<i>nui-dar-a(ba)</i>
<i>nui-dar-i</i>
<i>nug-ō</i>
<i>nug-e</i>
<i>nug-a-na-ku</i>
<i>nug-a-zu</i>
<i>nug-as-e</i>
<i>nug-ar-e</i>
<i>nug-u-mai</i>

<i>ut-</i>	<i>-Ø</i>	<i>-ru</i> [–PAST]
<i>ut-</i>	<i>-Ø</i>	<i>-reba</i>
<i>ut-</i>	<i>-Ø</i>	<i>#</i> [0 PAST]
<i>ut-</i>	<i>-Ø</i>	<i>#-tutu</i>
<i>ut-</i>	<i>-Ø</i>	<i>#-nagara</i>
<i>ut-</i>	<i>-Ø</i>	<i>#-gatera</i>

<i>ut-u</i> 'hit, beat'
<i>ut-eba</i>
<i>uti</i>
<i>uti-tutu</i>
<i>uti-nagara</i>
<i>uti-gatera</i>

<i>ut-</i>	<i>-Ø</i>	<i>#-ta-ku</i>
<i>ut-</i>	<i>-Ø</i>	<i>-te</i>
<i>ut-</i>	<i>-Ø</i>	<i>-temo</i>
<i>ut-</i>	<i>-Ø</i>	<i>-tatte</i>
<i>ut-</i>	<i>-Ø</i>	<i>-tar [+PAST]</i>
<i>ut-</i>	<i>-Ø</i>	<i>-tar-jō</i>
<i>ut-</i>	<i>-Ø</i>	<i>-tar-a(ba)</i>
<i>ut-</i>	<i>-Ø</i>	<i>-tar-i</i>
<i>ut-</i>	<i>-Ø</i>	<i>-jō</i>
<i>ut-</i>	<i>-Ø</i>	<i>-e</i>
<i>ut-</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>ut-</i>	<i>-Ø</i>	<i>-zu</i>
<i>ut-</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>ut-</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>ut-</i>	<i>-Ø</i>	<i>-mai</i>

<i>uti-ta-ku</i>
<i>ut-te</i>
<i>ut-temo</i>
<i>ut-tatte</i>
<i>ut-ta</i>
<i>ut-tar-ō</i>
<i>ut-tar-a(ba)</i>
<i>ut-tar-i</i>
<i>ut-ō</i>
<i>ut-e</i>
<i>ut-a-na-ku</i>
<i>ut-a-zu</i>
<i>ut-as-e</i>
<i>ut-ar-e</i>
<i>ut-u-mai</i>

<i>sin-</i>	<i>-Ø</i>	<i>-ru [-PAST]</i>
<i>sin-</i>	<i>-Ø</i>	<i>-reba</i>
<i>sin-</i>	<i>-Ø</i>	<i># [0 PAST]</i>
<i>sin-</i>	<i>-Ø</i>	<i>#-tutu</i>
<i>sin-</i>	<i>-Ø</i>	<i>#-nagara</i>
<i>sin-</i>	<i>-Ø</i>	<i>#-gatera</i>
<i>sin-</i>	<i>-Ø</i>	<i>#-ta-ku</i>
<i>sin-</i>	<i>-Ø</i>	<i>-te</i>
<i>sin-</i>	<i>-Ø</i>	<i>-temo</i>
<i>sin-</i>	<i>-Ø</i>	<i>-tatte</i>
<i>sin-</i>	<i>-Ø</i>	<i>-tar [+PAST]</i>
<i>sin-</i>	<i>-Ø</i>	<i>-tar-jō</i>
<i>sin-</i>	<i>-Ø</i>	<i>-tar-a(ba)</i>
<i>sin-</i>	<i>-Ø</i>	<i>-tar-i</i>
<i>sin-</i>	<i>-Ø</i>	<i>-jō</i>
<i>sin-</i>	<i>-Ø</i>	<i>-e</i>
<i>sin-</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>sin-</i>	<i>-Ø</i>	<i>-zu</i>
<i>sin-</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>sin-</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>sin-</i>	<i>-Ø</i>	<i>-mai</i>

<i>sin-u 'die'</i>
<i>sin-eba</i>
<i>sini</i>
<i>sini-tutu</i>
<i>sini-nagara</i>
<i>sini-gatera</i>
<i>sini-ta-ku</i>
<i>sin-de</i>
<i>sin-demo</i>
<i>sin-datte</i>
<i>sin-da</i>
<i>sin-dar-ō</i>
<i>sin-dar-a(ba)</i>
<i>sin-dar-i</i>
<i>sin-ō</i>
<i>sin-e</i>
<i>sin-a-na-ku</i>
<i>sin-a-zu</i>
<i>sin-as-e</i>
<i>sin-ar-e</i>
<i>sin-u-mai</i>

<i>kir-</i>	<i>-Ø</i>	<i>-ru [-PAST]</i>
<i>kir-</i>	<i>-Ø</i>	<i>-reba</i>
<i>kir-</i>	<i>-Ø</i>	<i># [0 PAST]</i>
<i>kir-</i>	<i>-Ø</i>	<i>#-tutu</i>
<i>kir-</i>	<i>-Ø</i>	<i>#-nagara</i>
<i>kir-</i>	<i>-Ø</i>	<i>#-gatera</i>
<i>kir-</i>	<i>-Ø</i>	<i>#-ta-ku</i>
<i>kir-</i>	<i>-Ø</i>	<i>-te</i>
<i>kir-</i>	<i>-Ø</i>	<i>-temo</i>
<i>kir-</i>	<i>-Ø</i>	<i>-tatte</i>
<i>kir-</i>	<i>-Ø</i>	<i>-tar [+PAST]</i>
<i>kir-</i>	<i>-Ø</i>	<i>-tar-jō</i>

<i>kir-u 'cut'</i>
<i>kir-eba</i>
<i>kiri</i>
<i>kiri-tutu</i>
<i>kiri-nagara</i>
<i>kiri-gatera</i>
<i>kiri-ta-ku</i>
<i>kit-te</i>
<i>kit-temo</i>
<i>kit-tatte</i>
<i>kit-ta</i>
<i>kit-tar-ō</i>

<i>kir-</i>	<i>-Ø</i>	<i>-tar-a(ba)</i>
<i>kir-</i>	<i>-Ø</i>	<i>-tar-i</i>
<i>kir-</i>	<i>-Ø</i>	<i>-jō</i>
<i>kir-</i>	<i>-Ø</i>	<i>-e</i>
<i>kir-</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>kir-</i>	<i>-Ø</i>	<i>-zu</i>
<i>kir-</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>kir-</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>kir-</i>	<i>-Ø</i>	<i>-mai</i>

<i>kit-tar-a(ba)</i>
<i>kit-tar-i</i>
<i>kir-ō</i>
<i>kir-e</i>
<i>kir-a-na-ku</i>
<i>kir-a-zu</i>
<i>kir-as-e</i>
<i>kir-ar-e</i>
<i>kir-u-mai</i>

<i>tob-</i>	<i>-Ø</i>	<i>-ru [-PAST]</i>
<i>tob-</i>	<i>-Ø</i>	<i>-reba</i>
<i>tob-</i>	<i>-Ø</i>	<i># [0 PAST]</i>
<i>tob-</i>	<i>-Ø</i>	<i>#-tutu</i>
<i>tob-</i>	<i>-Ø</i>	<i>#-nagara</i>
<i>tob-</i>	<i>-Ø</i>	<i>#-gatera</i>
<i>tob-</i>	<i>-Ø</i>	<i>#-ta-ku</i>
<i>tob-</i>	<i>-Ø</i>	<i>-te</i>
<i>tob-</i>	<i>-Ø</i>	<i>-temo</i>
<i>tob-</i>	<i>-Ø</i>	<i>-tatte</i>
<i>tob-</i>	<i>-Ø</i>	<i>-tar [+PAST]</i>
<i>tob-</i>	<i>-Ø</i>	<i>-tar-jō</i>
<i>tob-</i>	<i>-Ø</i>	<i>-tar-a(ba)</i>
<i>tob-</i>	<i>-Ø</i>	<i>-tar-i</i>
<i>tob-</i>	<i>-Ø</i>	<i>-jō</i>
<i>tob-</i>	<i>-Ø</i>	<i>-e</i>
<i>tob-</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>tob-</i>	<i>-Ø</i>	<i>-zu</i>
<i>tob-</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>tob-</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>tob-</i>	<i>-Ø</i>	<i>-mai</i>

<i>tob-u 'fly'</i>
<i>tob-eba</i>
<i>tobi</i>
<i>tobi-tutu</i>
<i>tobi-nagara</i>
<i>tobi-gatera</i>
<i>tobi-ta-ku</i>
<i>ton-de</i>
<i>ton-demo</i>
<i>ton-datte</i>
<i>ton-da</i>
<i>ton-dar-ō</i>
<i>ton-dar-a(ba)</i>
<i>ton-dar-i</i>
<i>tob-ō</i>
<i>tob-e</i>
<i>tob-a-na-ku</i>
<i>tob-a-zu</i>
<i>tob-as-e</i>
<i>tob-ar-e</i>
<i>tob-u-mai</i>

<i>jom-</i>	<i>-Ø</i>	<i>-ru [-PAST]</i>
<i>jom-</i>	<i>-Ø</i>	<i>-reba</i>
<i>jom-</i>	<i>-Ø</i>	<i># [0 PAST]</i>
<i>jom-</i>	<i>-Ø</i>	<i>#-tutu</i>
<i>jom-</i>	<i>-Ø</i>	<i>#-nagara</i>
<i>jom-</i>	<i>-Ø</i>	<i>#-gatera</i>
<i>jom-</i>	<i>-Ø</i>	<i>#-ta-ku</i>
<i>jom-</i>	<i>-Ø</i>	<i>-te</i>
<i>jom-</i>	<i>-Ø</i>	<i>-temo</i>
<i>jom-</i>	<i>-Ø</i>	<i>-tatte</i>
<i>jom-</i>	<i>-Ø</i>	<i>-tar [+PAST]</i>
<i>jom-</i>	<i>-Ø</i>	<i>-tar-jō</i>
<i>jom-</i>	<i>-Ø</i>	<i>-tar-a(ba)</i>
<i>jom-</i>	<i>-Ø</i>	<i>-tar-i</i>
<i>jom-</i>	<i>-Ø</i>	<i>-jō</i>
<i>jom-</i>	<i>-Ø</i>	<i>-e</i>
<i>jom-</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>jom-</i>	<i>-Ø</i>	<i>-zu</i>

<i>jom-u 'read'</i>
<i>jom-eba</i>
<i>jomi</i>
<i>jomi-tutu</i>
<i>jomi-nagara</i>
<i>jomi-gatera</i>
<i>jomi-ta-ku</i>
<i>jon-de</i>
<i>jon-demo</i>
<i>jon-datte</i>
<i>jon-da</i>
<i>jon-dar-ō</i>
<i>jon-dar-a(ba)</i>
<i>jon-dar-i</i>
<i>jom-ō</i>
<i>jom-e</i>
<i>jom-a-na-ku</i>
<i>jom-a-zu</i>

<i>jom-</i>	<i>-Ø</i>	<i>-sas-e</i>	<i>jom-as-e</i>
<i>jom-</i>	<i>-Ø</i>	<i>-rar-e</i>	<i>jom-ar-e</i>
<i>jom-</i>	<i>-Ø</i>	<i>-mai</i>	<i>jom-u-mai</i>

<i>kaw-</i>	<i>-Ø</i>	<i>-ru</i> [–PAST]	<i>ka-u</i> 'buy'
<i>kaw-</i>	<i>-Ø</i>	<i>-reba</i>	<i>ka-eba</i>
<i>kaw-</i>	<i>-Ø</i>	<i>#</i> [0 PAST]	<i>kai</i>
<i>kaw-</i>	<i>-Ø</i>	<i>#-tutu</i>	<i>kai-tutu</i>
<i>kaw-</i>	<i>-Ø</i>	<i>#-nagara</i>	<i>kai-nagara</i>
<i>kaw-</i>	<i>-Ø</i>	<i>#-gatera</i>	<i>kai-gatera</i>
<i>kaw-</i>	<i>-Ø</i>	<i>#-ta-ku</i>	<i>kai-ta-ku</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-te</i>	<i>kat-te</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-temo</i>	<i>kat-temo</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-tatte</i>	<i>kat-tatte</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-tar</i> [+PAST]	<i>kat-ta</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-tar-jō</i>	<i>kat-tar-ō</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-tar-a(ba)</i>	<i>kat-tar-a(ba)</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-tar-i</i>	<i>kat-tar-i</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-jō</i>	<i>ka-ō</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-e</i>	<i>ka-e</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-na-ku</i>	<i>kaw-a-na-ku</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-zu</i>	<i>kaw-a-zu</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-sas-e</i>	<i>kaw-as-e</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-rar-e</i>	<i>kaw-ar-e</i>
<i>kaw-</i>	<i>-Ø</i>	<i>-mai</i>	<i>ka-u-mai</i>

<i>tow-</i>	<i>-Ø</i>	<i>-ru</i> [–PAST]	<i>to-u</i> 'ask (question)'
<i>tow-</i>	<i>-Ø</i>	<i>-reba</i>	<i>to-eba</i>
<i>tow-</i>	<i>-Ø</i>	<i>#</i> [0 PAST]	<i>toi</i>
<i>tow-</i>	<i>-Ø</i>	<i>#-tutu</i>	<i>toi-tutu</i>
<i>tow-</i>	<i>-Ø</i>	<i>#-nagara</i>	<i>toi-nagara</i>
<i>tow-</i>	<i>-Ø</i>	<i>#-gatera</i>	<i>toi-gatera</i>
<i>tow-</i>	<i>-Ø</i>	<i>#-ta-ku</i>	<i>toi-ta-ku</i>
<i>tow-</i>	<i>-Ø</i>	<i>-te</i>	<i>tō-te</i>
<i>tow-</i>	<i>-Ø</i>	<i>-temo</i>	<i>tō-temo</i>
<i>tow-</i>	<i>-Ø</i>	<i>-tatte</i>	<i>tō-tatte</i>
<i>tow-</i>	<i>-Ø</i>	<i>-tar</i> [+PAST]	<i>tō-ta</i>
<i>tow-</i>	<i>-Ø</i>	<i>-tar-jō</i>	<i>tō-tar-ō</i>
<i>tow-</i>	<i>-Ø</i>	<i>-tar-a(ba)</i>	<i>tō-tar-a(ba)</i>
<i>tow-</i>	<i>-Ø</i>	<i>-tar-i</i>	<i>tō-tar-i</i>
<i>tow-</i>	<i>-Ø</i>	<i>-jō</i>	<i>to-ō</i>
<i>tow-</i>	<i>-Ø</i>	<i>-e</i>	<i>to-e</i>
<i>tow-</i>	<i>-Ø</i>	<i>-na-ku</i>	<i>tow-a-na-ku</i>
<i>tow-</i>	<i>-Ø</i>	<i>-zu</i>	<i>tow-a-zu</i>
<i>tow-</i>	<i>-Ø</i>	<i>-sas-e</i>	<i>tow-as-e</i>
<i>tow-</i>	<i>-Ø</i>	<i>-rar-e</i>	<i>tow-ar-e</i>
<i>tow-</i>	<i>-Ø</i>	<i>-mai</i>	<i>to-u-mai</i>

The interpretation of the morphophonological operations described above is as follows.

1. Insertion of -ru before -mai

Before the negative-volitional/assumptive suffix *-mai*, consonant-stem verbs show insertion of the non-past tense suffix *-ru* following the V thematic suffix *-Ø*. Thus:

kes-Ø-mai → *kes-Ø-**ru**-mai* (→ *kes-ru-mai* → *kes-u-mai*)

This process indicates that *-mai* does not attach directly to the bare consonant stem. Rather, the non-past suffix *-ru* is first introduced, after which the regular morphophonological operations apply. In the *s*-irregular verb *s-u-ru* 'do' and the *k*-irregular verb *k-u-ru* 'come', insertion of *-ru* following the V thematic suffix *-i* is also optionally possible. Thus:

s-i-mai → *s-i-**ru**-mai* (→ *s-u-ru-mai*)

k-i-mai → *k-i-**ru**-mai* (→ *k-u-ru-mai*)

By contrast, the other vowel-stem verbs canonically do not show this insertion. Thus:

tab-e-mai → *tab-e-mai*, but not **tab-e-**ru**-mai*

2. Stem alternations

Before the negative suffixes *-na** and *-zu*, the V thematic suffix *-Ø* alternates with *-a*, yielding a special stem used only before the negative suffixes. Thus:

kes-Ø-na-ku → *kes-**a**-na-ku*

kes-Ø-zu → *kes-**a**-zu*

In the *s*-irregular verb *s-u-ru* and the *k*-irregular verb *k-u-ru*, the following alternations of the V thematic suffix are attested:

s-i → *s-**u*** / ___ *-ru*, *-reba*

s-i → *s-**e*** / ___ *-zu*, *-jo*, *-sas**, *-rar**

k-i → *k-**u*** / ___ *-ru*, *-reba*

k-i → *k-**o*** / ___ *-jō*, *-jo*, *-na**, *-zu*, *-sas**, *-rar**, *-mai*

In the *s*-irregular verb, moreover, the following alternation of the V thematic suffix *-i* is possible:

$$s\text{-}i \rightarrow s\text{-}u / __ -mai$$

Lastly, in the vowel-stem verbs *Ø-e-ru* 'gain' and *p-e-ru* [heru] 'pass (time)', the following alternations of the V thematic suffix are possible in literary register:

$$\text{Ø}\text{-}e \rightarrow \text{Ø}\text{-}u / __ -ru, -reba$$

$$p\text{-}e \rightarrow p\text{-}u / __ -ru, -reba$$

3. Deletion of the remaining V thematic suffix -Ø

Any remaining *-Ø* is deleted. In other words, except in the environments specified in 2, the V thematic suffix *-Ø* has no overt phonological realisation. Thus:

$$kes\text{-}\text{Ø}\text{-}ru \rightarrow kes\text{-}ru$$

$$kes\text{-}\text{Ø}\text{-}ru\text{-}mai \rightarrow kes\text{-}ru\text{-}mai$$

4. Deletion of suffix-initial consonants

Suffix-initial *s*, *r* and *j* are deleted after any consonant, whether stem-final or suffix-final. Thus:

$$kes\text{-}ru \rightarrow kes\text{-}u$$

$$kes\text{-}ru\text{-}mai \rightarrow kes\text{-}u\text{-}mai$$

$$kes\text{-}reba \rightarrow kes\text{-}eba$$

$$kes\text{-}tar\text{-}j\bar{o} \rightarrow kes\text{-}tar\text{-}\bar{o} (\rightarrow kes\text{-}i\text{-}tar\text{-}\bar{o})$$

$$kes\text{-}j\bar{o} \rightarrow kes\text{-}\bar{o}$$

$$kes\text{-}sas\text{-}e \rightarrow kes\text{-}as\text{-}e$$

5. Voicing in the *t*-series suffixes

In the suffixes *-te*, *-temo*, *-tatte*, *-tar**, suffix-initial *t* becomes *d* after stem-final *b*, *g*, *m*, *n*, that is, after consonants specified as [−sonorant, +voiced] or [+nasal]. Thus:

-bt-, -gt-, -mt-, -nt-
 → -bd-, -gd-, -md-, -nd-

These suffixes therefore form a class that attaches directly to the consonant-final stem and triggers immediate morphophonological interaction with it. Thus:

nug-te → *nug-de*
sin-te → *sin-de*
tob-te → *tob-de*
jom-te → *jom-de*

6. Alternations before the d-series allomorphs of the t-series suffixes

Before the *d*-series allomorphs *-de*, *-demo*, *-datte*, *-dar** of the *t*-series suffixes, stem-final *b* and *m* become *n*. This alternation may be understood as the final phase in a process involving nasalisation followed by place assimilation. Before the same allomorphs, by contrast, *g* surfaces as *i*. More precisely, this development is analysed not as a direct change *g* → *i*, but as a process involving insertion of *i* followed by deletion of *g*. The sequence may be represented as follows:

-bt-, -mt-, -nt-, -gt-
 → -bd-, -md-, -nd-, -gd-
 → -**md**-, -md-, -nd-, -gd-
 → -**nd**-, -**nd**-, -nd-, -gd-
 → -nd-, -nd-, -nd-, -**gid**-
 → -nd-, -nd-, -nd-, -**id**-

The interpretation is as follows. First, *t* becomes voiced after a nasal or a voiced obstruent. Next, *b* becomes *m* in order to avoid a sequence of voiced obstruents. Then, *m* becomes *n* by place assimilation to *d*. In the case of *g*, nasalisation to *ŋ* is excluded — possibly by structure preservation — and assimilation to *d* is likewise excluded. The cluster is therefore broken up by insertion of *i*, after which *g* is deleted. Thus:

tob-de → *tom-de* → *ton-de*

jom-de → *jon-de*

nug-de → *nugi-de* → *nui-de*

7. Coalescence before the *t*-series suffixes

Before *-te*, *-temo*, *-tatte*, *-tar**, *ow* becomes *ō*. Thus:

tow-te → *tō-te*

8. Assimilation and insertion before the *t*-series suffixes

Before *-te*, *-temo*, *-tatte*, *-tar**, stem-final *r* and *w* assimilate to *t*. Thus:

-rt-, *-wt-* → *-tt-*, *-tt-*

Before the same suffixes, *s* and *k* surfaces as *si* and *ki* rather than assimilating to *t*. Accordingly:

-st-, *-kt-* → *-sit-*, *-kit-* → *-sit-*, *-it-*

This is because the assimilation *-st-*, *-kt-* → *-tt-*, *-tt-* is not permitted, except in the verb *ik-u* 'go', underlyingly *ik-Ø-ru*, where *-kt-* → *-tt-* is observed. Instead, *i* is inserted after *s* and *k*, yielding the sequences *si* and *ki*, after which *k* is deleted. The overall pattern may be stated as follows:

-tt-, *-st-*, *-rt-*, *-kt-*, *-wt-*
→ *-tt-*, *-st-*, *-tt-*, *-kt-*, *-tt-*
→ *-tt-*, *-sit-*, *-tt-*, *-kit-*, *-tt-*
→ *-tt-*, *-sit-*, *-tt-*, *-it-*, *-tt-*

Accordingly:

ut-te → *ut-te*

kes-te → *kesi-te*

kir-te → *kit-te*

kak-te → *kaki-te* → *kai-te*

ik-te → *it-te*

kaw-te → *kat-te*

9. Insertion of *i* before the morpheme boundary

Before the morpheme boundary #, *i* is inserted after any stem-final consonant. This process does not need to refer separately to *-tutu*, *-nagara*, *-gatera*, *-ta**. Those forms are analysed as occurring outside this same boundary. Accordingly, the relevant environment is not the initial consonant of the following suffix, but the boundary # itself. Thus:

ut# → *uti#*

ut#-tutu → *uti#-tutu*

ut#-nagara → *uti#-nagara*

ut#-gatera → *uti#-gatera*

ut#-ta-ku → *uti#-ta-ku*

The insertion of *i* therefore applies uniformly before #, regardless of whether further material follows outside that boundary. In this sense, the stem form seen before *#-tutu*, *#-nagara*, *#-gatera* and *#-ta** is the same as that seen before # alone.

10. Deletion of *w* outside *a**

The consonant *w* is deleted except before *a**. Thus:

kaw-u → *ka-u*

kaw-ō → *ka-ō*

tow-i → *to-i*

tow-eba → *to-eba*

By contrast, *w* is retained before *a**. Thus:

kaw-a-na-ku → *kaw-a-na-ku*

tow-ar-e → *tow-ar-e*

This is enforced by the exclusion of Yamato and Sino-Japanese **wi*, **we*, **wo*, **wu* sequences.

11. Deletion of word-final *r*

Word-final *r* is deleted. Thus:

kesi-tar → *kesi-ta*

This operation eliminates a final consonant not permitted on the surface.

12. Allomorphy of *-jo* as *-i*

The imperative suffix *-jo* is realised as *-i* only in the ***k-irregular*** verb. Thus:

ko-jo → *ko-i*

13. Short structural interpretation

The system therefore distinguishes two suffix classes. The first class consists of suffixes such as *-te*, *-temo*, *-tatte*, *-tar**, which attach directly to the consonant-final stem and trigger immediate alternations such as voicing, assimilation and vowel insertion. The second class consists of forms such as *-tutu*, *-nagara*, *-gatera*, *-ta**, which do not interact directly with the stem-final consonant. Instead, they are introduced outside the morpheme boundary #. As a result, the stem first undergoes the same process that appears before # alone, namely insertion of *i* after a stem-final consonant. Accordingly, the opposition is not simply between two sets of consonant-initial suffixes. Rather, it is a contrast between suffixes that attach directly to the stem and suffixes that are introduced only outside the boundary #.

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